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KENYATTA UNIVERSITY

INSTITUTE OF OPEN LEARNING

DEPARTMENT OF GEOGRAPHY

**AGE 111: DEVELOPMENT OF GEOGRAPHIC
THOUGHT**

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INTRODUCTION TO THE COURSE

AGE 101 (Development of Geographic Thought) is an important foundation course that introduces students to some fundamental concepts in the development of Geography. The course specifically traces the evolution of Geography as a distinct discipline from the ancient times, through the middle ages and the Renaissance period to the classical and post-classical periods. The course is highly philosophical and thought provoking. He/she who properly grasps the concepts and contributions of various personalities to the development of Geographic thought inevitably becomes a good student of Geography. I am optimistic that you will find the course relevant to your future academic and professional needs.

You are expected to read the module very carefully and do all the reading and written assignments that are given in the module. I urge you to read and discuss extensively in this course and share your knowledge and any references that you get with your friends.

LECTURES 1,2 & 3

DEFINITION, ORIGIN AND SCOPE OF GEOGRAPHY.

Introduction:

In these first three lessons, I will attempt to give a definition of the discipline geography.

The origins of geography will also be briefly discussed noting its origin from the Greeks.

The lessons will finally look at the scope of modern geography.

Objectives:

At the end of these lessons, you should be able to:-

- a) Define the term Geography
- b) Briefly discuss the origin of Geography and
- c) Describe the concerns of contemporary Geography.

Origin of Geography.

The word 'geography' is derived from two Greek words Geo (meaning earth) and graphy (meaning writing or describing). In literal terms, geography therefore means earth writing, or writing about the earth. A Greek philosopher called Eratosthenes who worked in the museum at Alexandria is widely believed to have coined the term geography.

Geography is primarily concerned with the relationship between man's way of life and the places in which he lives. The aim of geography is to understand the earth as the world of man, with particular reference to the differentiation and integration of places. Over the last

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3,000 years or so geography has been concerned with accurately describing the

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location of places. This is an activity whose origin can be traced as far back as the early Greek civilization. But this interest in the location of places alone cannot be regarded as the actual origin of the subject matter.

Question

Why should people study geography?

What is Geography?

Most people have very vague notions about the content of scientific geography. School geography has left many with bad memories of learning the names of rivers and towns by rote memorization. It is still common to meet people who think that geographers must have to learn a mass of facts, must know the population of towns all over the world and can name and locate all the major rivers of the world. This idea of geography as the encyclopedic knowledge of places is a notion that is still widely held by most people.

Other people have the idea that geography has something to do with maps. Geographers are thought to be people who know how to draw maps. A third view is that geographers write travel descriptions. Each of these popular opinions as to what geography is has some truth in it. The names and locations of towns are facts for geographers of the same order as dates are facts for historians. They are the basic building blocks of the subject, but they are not the subject itself. The map, which represents a collection of such data, is the geographer's most useful specific resource. Different types of thematic maps are also

important means of expression in geographical research, along with tables, diagrams and

written accounts. Observations recorded during travel and fieldwork still provide essential data for geographers. A cultivation of the powers of observation is therefore an important objective in the education of a geographer. It is of primary importance to learn how to 'see geographically' - to observe and to interpret a natural or cultural landscape without having to have local knowledge of it. Geographical training develops a distinct method of observation aimed at the understanding of the form of culture and landscape in quite a different way from that of the ordinary tourist.

Contemporary Definition of Geography.

Geography is concerned with providing accurate, orderly and rational description and interpretation of the variable character of the earth's surface. A traditionally held view - that geography is concerned with giving man an orderly description of his world - makes clear the challenge faced by contemporary geographers. The contemporary emphasis is on geography as the study of spatial organization expressed as patterns and processes. Geography can be regarded as a science concerned with the rational development and testing of theories that explain and predict the spatial organization expressed as patterns and processes. Geography can be regarded as a science concerned with the rational development and testing of theories that explain and predict the spatial distribution and location of various characteristics on the surface of the earth.

Prominent Geographers who have immensely contributed to geographic thought (such as Hartshorne and Yeates) differ on the methods rather than the content of Geography. While

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others such as Yeates and Hartshorne are concerned with the variation of

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phenomena over the surface of the earth, Hartshorne visualizes geography as an idiographic science with its main emphasis on the description and elucidation of individual phenomena because they are unique. Yeates considers geography to be a nomothetic (nomological) or law seeking science which requires the development and testing of theories and models through hypothetic - deductive methods in order to develop geographic laws.

Taaffe sees geography as a dynamic science having changed from an idiographic to a nomothetic science. Explanatory models, which were once thought to be satisfactory in explaining spatial variations of phenomena, were discredited by a large number of geographers. Adams sees geography as being in the throes of a paradigm crisis. Instead of asking the traditional question, "Is it Geography?" or "what is Geography?" Geographers are now asking "What should geography be?" If a satisfactory answer is not found to the latter question the next question is likely to be "Is Geography relevant?"

Today discoveries in outer space make the greatest appeal to man's urge to discover the unknown. Space research, however, requires highly specialized techniques than geographers or geographical societies can command. The collection and mapping of spatial data has been taken over by government organizations. The collection of geo referenced data using G.I.S. techniques has led to an explosion of information about the world so that the knowledge available to the modern geographer is many hundred times greater than his predecessor in the mid 19th century. Further development of G.I.S. and remotely sensed data will continue to multiply the amount of information. Electronic

computers and advanced statistical techniques have become essential for the handling of the growing flood of information about geographical phenomena. The use of these techniques is not, of course, confined to geography but is common to all sciences. For a science to play a distinctive and useful role, it needs both a specific field of enquiry and a set of specific concepts.

Activity

Write an essay on the circumference of Geography showing its relationship with other subjects.

Note

Geography may be regarded as a science of synthesis. Geography has no obvious place in the traditional classification of sciences by faculty. Some parts of geography have their strongest affiliations with mathematics and natural sciences. Other sciences study distinctive types of phenomena: geologists study rocks, botanists study plants, sociologists social groups, geomorphologists study landforms etc. The work of geographers involves the study of several types of phenomena, each already studied by another science.

Question.

Are Geographers jacks of all trades and masters of none or is geography a mother discipline

Note

It should further be noted that the subject matter of geography is shared with other disciplines but is treated in a different way for geographical purposes. Some geographers affirm that the subject matter of geography is exclusive; geographers alone study places. However, the fundamental approach in geography is the differentiation of the content of space on the earth's surface and the analysis of space relations within the same universe. Other geographers argue that geography should cultivate its core, which is regional geography, as a shock absorber against absorption by other sciences. Yet others see geography as a mother discipline from which others see geography as a mother discipline from which other specialized disciplines such as meteorology, soil science, plant ecology, and regional science have emerged. Geography has become an outward looking discipline that has frequently created new specializations. This multi-disciplinary perspective may be regarded both as geography's root and life raft in the sea of knowledge.

What do you consider as the major concerns of contemporary geography?

Activity**Summary**

In these first three lectures, we have looked at the origin, definition and concerns of geography. We have noted that the word geography has its roots from the Greek philosopher, Eratosthenes. We have also noted that most people have very vague notions

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about the content of scientific geography. We have finally looked at the concerns of

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modern geography noting that contemporary geographers emphasize on geography as the study of spatial organization expressed as patterns and processes. We have seen that geography can be regarded as a science concerned with the rational development and testing of theories that explain and predict the spatial distribution and location of various characteristics on the surface of the earth.

Definition of key words and concepts.

- Geography: An academic discipline that owes its origin to the ancient Greeks. Today geography is a discipline that emphasizes the study of spatial organization expressed as patterns and processes.
- Eratosthenes: A Greek scholar at the museum in Alexandria who is widely regarded as the first to coin the term geography.
- Geomorphologists: People who study landforms.
- Regional Geography: Geographical study of areas in their total composition and complexity.

Further reading

1. Unwin, T. (1994). **The place of Geography**, New York: Longman Scientific & Technical.
2. Holt - Jensen, A. (1982). **Geography: its History and concepts**, London: Harper and Row.
3. Buttimer, A. (ed.) (1983). **The practice of Geography**: London, Longman.

4. Chorley, R. J. and Hagget, P. (eds.) (1967). **Models in Geography**. London, Methuer.

LECTURES 4,5 & 6

GEOGRAPHY IN THE ANCIENT WORLD: GREEK AND ROMAN GEOGRAPHY.

Introduction:

In the previous lessons, we have looked at the origin and definition of geography. We have seen that the term geography is derived from two Greek words geo (meaning earth and graphy (meaning description). We have also noted the contemporary definition of geography as a discipline that emphasizes spatial organization of phenomena.

In the next three lectures, we are going to look at geography in the ancient world. In particular, we shall look at the contributions of the Greeks and Romans to the development of geographic thought during these ancient times.

Objectives.

By the end of these lectures, you should be able to :

- a) explain the major contribution of ancient Greeks to the development of Geography,
- b) Describe the contributions of Herodotus, Eratosthenes, Ptolemy and Posidonius to the development of geographic thought.
- c) Identify the contributions of Pythagoras, Strabo, Plato and Aristotle to the development of geography.

Contributions of Greeks and Romans to the Development of Geography.

Interest in geographical problems and writings on subjects that we can recognize as geographical, began long before the introduction of the subject into Universities. In fact it is difficult to imagine how there were ever people who did not think geographically, who never considered the conditions under which they lived, and never wondered how people lived in other places. In this sense, geographical thinking is older than the term 'Geography', which as we have noted was first used by scholars at the museum in Alexandria about 300 B.C.

The ancient Greeks made the first major contribution to the development of Geography. As noted earlier, geography literally means earth-writing or writing about the earth. However, the production of books specifically called geographies was a relatively late phenomenon, and until the 1st century B.C. most accounts of the earth were written in works that did not explicitly claim to be geographical. Before the advent of writing there was an informal pre-geography created as a result of the explorations and discoveries of prehistoric peoples. This led to the establishment of a body of knowledge about the world, which gradually formed the basis of the formal pre-geography of the classical world. In early Greek writing three such traditions can be identified:

- i) A topographical tradition concerned with descriptions of the earth and the people living on it;

- ii) A mathematical and astronomical tradition concerned with measurement of the earth;
- iii) A theological tradition concerned with answering questions about the very reason for human existence on the earth.

Question.

What utilitarian purpose does geographical knowledge serve today?

The topographical tradition of geographical writing was derived directly from travels of pre-literate peoples. As commercial exchange became firmly established it was essential for people to have knowledge of sea and land routes, and as explorers discovered new places it was important for knowledge about them to be disseminated within their societies. From its earliest days, therefore, geographical knowledge can be seen to have served a utilitarian purpose, and maps sketched on rocks and bones survive in the archaeological record from as early as 13,000 B.C. (Harvey, 1980). With the advent of writing such knowledge became formalized, and many of the earliest surviving poems and literary works provide considerable details about different parts of the then known world. Homer's Iliad and Odyssey (9th Century B.C) are thus widely considered to be among the first surviving geographical works. These accounts of the epic conflict between the Achaeans and the Trojans, and the subsequent wanderings of Odysseus, provided Homer with a marvelous back drop against which he could develop detailed descriptions of the people he described, and the places to which they traveled. The subsequent exploration and

colonization of much of the Mediterranean by the Greek city states in the 8th and 7th centuries BC led to a vast expansion of knowledge, much of which

was written down in treatises which have not survived. It is also from this period that the first serious attempts at representing the known world cartographically were attempted with Anaximander of Miletus (611-547 BC) widely being recognized as the author of the first map of the earth about which we have any real knowledge was the *Periodus*, or Description of the earth, completed by Hecataeus of Miletus (c. 550-476 BC), and divided into two books entitled Europe and Asia. Unfortunately only about 300 fragments of these works survive, and although Hecataeus is most usually considered to be a historian (Pearson, 1939), Thomson (1948:47) suggest that in his work 'there are hints of intelligent curiosity about climate and customs, flora and fauna, so that it deserves to be called a general geography.

Activity

Briefly describe the contributions of Anaximander and Hecataeus to the development of geographic thought
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All of this earlier topographical writing culminated in the first great prose work of European literature, the histories of Herodotus (c. 485-425 BC) (1954). In this, the result of what he termed his enquiries into history, Herodotus (1954:13) aimed 'to do two things: to preserve the memory of the past by putting on record the astonishing achievements both of our own and of the Asiatic peoples; secondly, and more particularly, to show how the two races came into conflict. In so doing, he provided a wide-ranging account of the whole of the known world, based on his own experiences on existing inscriptions and literary works, and on oral accounts. Although many of his stories now seem fantastic, much of

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Herodotus's account has been substantiated by other

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literary and archaeological evidence. In Book Four, for example, although he had little knowledge about the interior of Africa, which he termed Libya, he was aware that it was washed on all sides by the sea except where it joins Asia (Herodotus, 1954: 254). Likewise, in Book One he noted with surprise how Babylon was supplied with wine by boats made of hide, which were taken to pieces on arrival and the hide then carried by donkey overland back to Armenia since it is quite impossible to paddle the boats upstream because of the strength of the current' (Herodotus, 1954:92). A century after the death of Herodotus, Alexander the great (356-323 BC) vastly expanded the horizons of the Greek world with his eastern conquests of the Persian Empire. Although Herodotus had written much about Persia, these conquests provided the opportunity for a considerable increase in the accuracy of Greek topographical writings about the region. These found their fruition in the subsequent writings of Dicaearchus, a pupil of Aristotle, and of Eratosthenes (c. 276-194BC).

Activity

Enumerate the contributions of Herodotus to the development of geographical thought.
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At the same time as a topographical awareness of the earth's surface was growing, there also developed an interest in its measurement and in astronomy. To the Greeks of Homer's age, the earth was envisaged as a circular plane surrounded on all sides by an ocean. However, by the 6th century, a growing interest in astronomy was leading to a reappraisal of this view. Thales of Miletus (fl. 580 BC), one of the founders of the Ionian school of

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philosophy, for example, is widely regarded as having understood the cause of

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solar eclipses (Dreyer, 1953.) Thus Herodotus (1954:42) commented that Thales had predicted an eclipse during the battle between the Medes and the Lydians and Plutarch claimed that he had attributed solar eclipses to the interposition of the moon between the earth and the sun. However, there is no direct indication that he considered that the earth was a sphere. Anaximander is thought to have conceived of the earth as a cylinder, but it is once again only in later commentaries, particularly through the works of Aristotle (384-322 BC) that his ideas have survived. In essence, Pythagoras is claimed to have argued that the earth, along with the other visible heavenly bodies, revolved around a central fire, which occupied the mid-point of the universe. Since the sphere and the circle are the most perfect forms, according to Pythagoras, then the heavenly bodies, including the earth, must all be spheres revolving in circles. It was thus by somewhat mystical and philosophical, rather than strictly empirical, reasoning that it was argued that the earth was round.

The description of the earth as a sphere by Plato (c. 427-347 BC) in *the republic*, and in *Timaeus* was to have a marked significance for the future understanding of its shape. Indeed, Thomson (1948) has argued that 'it was Plato's adoption that gave the globe a wider currency'. However, by placing the spherical earth at the center of the universe, with the sun, moon and planets orbiting around it, Plato's account was to hamper deeper astronomical understanding for centuries to come. In contrast to Plato's concern with the unseen world of ideal forms, Aristotle's emphasis on the collection of empirical facts was eventually to lead to a greater concern with the measurement of the earth's circumference. These ideas were taken up by his pupils, notably Dicaearchus, and subsequently reached

their fruition in the research undertaken at Alexandria, particularly under Eratosthenes, who is generally acknowledged to have made the first accurate measurement of the globe and who has been described by Bunbury (1879;) as 'the parent of scientific geography'. Eratosthenes, moreover, also developed a system of latitude and longitude, and published a three volume treatise entitled *Geographica*, which, although it does not survive in its original form, was one of the main sources used by Strabo and Ptolemy.

Activity

Explain the contributions of Thales, pythagoras, Plato and Eratosthenes to the development of geography

These topographical and astronomical accounts were also closely associated with the third, theological, tradition concerning the origins of the earth and the reasons for human existence on it. Indeed, they can also be seen as combining together to an interest in astrology, the use of astronomy to predict human and natural events on the earth. Glacken (1967: 5) has suggested that three basic ideas underlay much early mythology concerning the relationships between people and the environment: that the earth` had an order and purpose designed by a deity; that the environment had an influence on people; and that people were able to modify the environment. Gradually the Ionian school of philosophers began to develop a particular kind of cosmology, or theory of the composition of the universe. Thales of Miletus had suggested that everything was made of water, but this theory was subsequently revised by Anaximander who envisaged a primal infinite substance from which the many substances of the world, including water, were derived

(Kahn, 1960; Russell, 1961). At the same time, Anaximander also developed an argument

that 'Order is characterized by a struggle of opposites' (Glacken, 1967:9). Eventually, in the 5th century BC the influential theory of the four roots or elements, was formulated by Empedocles of Acragas (Agrigentum) in southern Sicily (c. 492-432 BC). His major contribution to science was to identify air as a separate substance. This then enabled him to suggest that there were four primitive independent elements, earth, air, fire and water, which could be combined in different proportions to produce the many different substances found in the world. These elements were held together by love, and separated by hate, thus representing a further refinement of Anaximander's theory of opposites. It was not until the 4th century that Aristotle added a fifth element, ether (On the heavens). For Aristotle, the spherical earth was at the center of the universe. Everything below the moon was considered to be made of the four elements, subject to change and decay, whereas everything above the moon, was made of the fifth element, ether, which was indestructible.

Activity

Itemize the major contributions to geographic thought by Empedocles and Aristotle

Meanwhile, alongside this emerging cosmology, there developed a range of ideas concerning the place of people within the global order. These differed markedly, and as Glacken (1967: 13) has pointed out. 'The single most important generalization to be made about the attitudes toward nature held by the peoples of the classical world is that these varied greatly throughout the long span of history'. Underlying the classical vision, though, lay much older religious concepts, the most important of which were those

associated with fertility, death and rebirth. At the heart of this mythology lay the worship of a female, mother earth, later to be formalized as the great deity of the primitive Greeks, Gaea (Guirand, 1968). Even before the origins of agriculture, human survival depended on the annual cycle of the plants, and on the fertility of nature. The fertility of the natural world was the same as that of the human world, and human ritual sought to propitiate a bountiful mother earth, frequently through the sacrifice of male animals. This concept of a female earth was not, though, universal, for in Egyptian mythology it is salient to note that the earth-god known as *Geb* was male, in contrast to the female goddess of the sky, *Nut* or *Hathor* (Viaud, 1968).

The idea of a planned world subject to divine intervention is extremely ancient, and hints of it can be found in the early myths of Egypt, and the peoples of the Tigris-Euphrates basin. By the 5th century, though, a firm teleology, or doctrine concerning design and purpose in nature, had emerged at Athens. Thus Xenophon (c.435-354 BC) in his *Memorabilia* describes Socrates (before 469-399 BC) as using three main types of evidence to demonstrate 'divine providence: the proof of physiology, of the cosmic order, and of the earth as a fit environment' (Glacken, 1967:42). Plato developed some of these ideas in his *Timaeus*, where he provides an account of the framer of the universe, creating the ordered cosmos consisting of fire, earth, air and water out of a state of disorder (*Timaeus* 29-30). In particular, Plato suggested that god was good, and therefore that he designed his ordered creation so that 'all things should be good, and so far as possible nothing be imperfect' (Plato, 1971: 42). Aristotle appears to have abandoned the idea of a creator god, and instead

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envisaged an internal logic in nature, which drove it forward of

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necessity (Physics); nature acts for a purpose, but for a purpose that is unconscious. Such arguments, particularly those of Socrates, had a major influence on the Stoic school of philosophy, established by Zeno in the early part of the 3rd century BC, and through them on much subsequent European philosophy. Zeno believed that the whole of nature was determined by rigorous laws, designed for the benefit of humanity. Russell (1961: 262) thus argues that for the Stoics 'everything has a purpose connected with human beings. Some animals are good to eat, some afford tests of courage; even bed bugs are useful, since they help us to wake in the morning and not lie in bed too long.' Such views did not, however, go unquestioned, and found their most powerful opposition in the school of philosophy established by Epicurus (c. 341-270 BC). The Epicureans, including Cicero (106-43 BC) and Lucretius (c. 99-55BC), pursued pleasure as the main good, and conceiving that fear of death and fear of the gods were the main evils, they were thus forced to separate the gods from the main evils, they were thus forced to separate the gods from nature. Given that not everyone living on the earth is fully content, they argued that nature cannot have been designed purely for human beings.

Contribution of Strabo and Ptolemy to the development of geographic thought.

By the end of the Hellenistic age, and the emergence of Rome as the dominant Mediterranean power, geography was considered to encompass not only topographical descriptions of the world, but also questions of astronomy and teleology; right at the heart of classical geography lay profound philosophical questions concerning the place of people

in nature. Its subject matter was closely similar, if not necessarily interchangeable,

with that of physics literally meaning 'natural things', and geometry meaning 'earth - measuring' Earth-writing (geography) thus included work on the realm of natural things, the measurement of the earth, and theological matters concerning the role of divine power in forming the earth.

Specific works on geography, however, remained few in number. While authors such as Polybius (c. 205-c. 123 BC) and Posidonius (c. 135-52 BC) included much topographical description in their works, they described them as histories rather than geographies. What geographical works that were written have generally failed to survive. It is thus with the Geography of Strabo (c. 60. BC - post AD 21), a Greek born at Amasia in Pontus, that a formal geography can finally be considered to have found its surviving expression. Significantly, Strabo did not consider himself to be purely a geographer, and if his forty seven book history had survived he might well have been better known as a historian. Nevertheless Strabo's work does enable us, in Bunbury's (1879) words, for the first time to obtain a complete and satisfactory view of the state of geographical science.

More importantly, Strabo also provides a justification for the enterprise of writing geography. He begins the first of the seventeen books of his Geography by arguing that the science of Geography, which I now propose to investigate, is, I think, quite as much as any other science, a concern of the philosopher' (Strabo, 1949; 3). He asserts this on the grounds.

1. That those who had previously discussed the subject, such as Homer and Anaximander,

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were philosophers;

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2. That 'wide learning' which alone makes it possible to undertake a work on geography, is possessed solely by the man who has investigated things both human and divine - knowledge of which, they say, constitutes philosophy' (Strabo, 1949:3).
3. That the utility of geography 'presupposes in the geographer the same philosopher, the man who busies himself with the investigation of the art of life, that is, of happiness' (Strabo, 1949:5).

Nowhere does Strabo provide a clear and straightforward definition of geography, but within Book One he claims that it requires encyclopedic learning and also that 'All those who undertake to describe the distinguishing features of countries devote special attention to astronomy and geometry' (Book 1. 1. 12-13) (Strabo, 1949:25). Moreover to this he adds terrestrial history, which he defines as the history of animals, plants and everything produced by land or sea (Book 1.1. 16), and also a considerable emphasis on the political development of states (Book 1.1. 18). In addition to these empirical aspects, he claims that geography involves theory of no mean value, the theory of the arts of mathematics, and of natural sciences, as well as the theory which lies in the fields of history and myths' (Book 1.1. 19) (Strabo, 1949:39).

It is in Strabo's account of the utility of geography; though, that we can gain a real insight into the role that the discipline played in Greek and Roman society. Geography's central task was seen to be political.

Strabo thus argues that the greater part of geography subserves the needs of states and that geography as a whole has a direct bearing upon the activities of commanders' (Book 1.1. 16) (Strabo, 1949: 31). Indeed, he asserts that the greatest generals are without exception men who are able to hold sway over land and sea, and to unite nations under one government and political administration' (Book 1.1. 16) (Strabo, 1949:31). The role of geography was thus to provide the information that enabled rulers to conquer more territory, and to retain power over the lands that they rule; the description which geography gives is of importance to these men who are concerned as to whether this or that is so or otherwise, and whether known or unknown. For thus they can manage their various affairs in a more satisfactory manner, if they know how large a country is, how it lies, and what are the peculiarities either of sky or soil' (Book 1.1. 16) (Strabo, 1949:33). Geography is, though, not only important in great undertakings, and Strabo uses the illustration of hunting to emphasize its importance for more successful in the chase if he knows the character and extent of the forest; and again, only one who knows a region can advantageously pitch camp there, or set an ambush, or direct a march' (Strabo, 1949; 35).

In terms of content, Strabo freely admits that he has made use of the works of his predecessors, notably Eratosthenes, Polybius and Aristotle, and indeed it is primarily through Strabo's writing that there remains any detailed knowledge of many earlier geographical works at all. He assumes that the earth is spherical, lying in the center of the universe, and divided into five zones; its inhabited portion is described as a large island, surrounded on all sides by the ocean. The bulk of his geography, though, consists mainly

of a topographical description of the known world and after the first two introductory books, eight books are allocated to Europe, six to Asia, and one to Africa.

The next major description of the known world after Strabo's Geography was the Natural history of Pliny (AD 23-79) (Pliny, 1855-57). Although this did not specifically claim to be a work of geography, it aimed to present an overview of everything that was then known of the physical constitution of the universe. Pliny's work has been subjected to considerable criticism, largely on the grounds of its lack of structure and poor scientific understanding. Indeed Bunbry (1879: 34) has argued that when Pliny's geographical writings are compared with the writings of Eratosthenes and Strabo, we are struck with the almost total absence of any scientific comprehension of his subject, and of those general views which, however imperfectly developed, were certainly present in the minds of the Greek geographers'. The problems with Pliny's Natural history derive in part from its design and method of compilation. In his preface (Book 1) Pliny thus claimed that he had included 20,000 topics gained from the perusal of 2,000 volumes written by some 100 authors (Pliny, 1855:1.7), and that these were put together at interrupted intervals. It is therefore not surprising that there is a certain amount of confusion and disorder in his work. Book II of the Natural history is 'An account of the world and the elements', which sets the globe within its astronomical setting. This is followed by four books on geography which Pliny describes as containing 'an account of the situation of the different countries, the inhabitants, the seas, towns, harbors, mountains, rivers, and dimensions, and the various tribes, some of which still exist and others have disappeared' (Book 1) (Pliny, 1855:1.11). by making a

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break between his astronomical and

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geographical observations, Pliny therefore appears to be adopting merely a topographical definition of geography. However, it is interesting to note that by discussing tribes that have disappeared he is able to include a historical dimension in his geographical writing. In the remaining books, Pliny describes man and his inventions, animals, birds. Plants, medicines and minerals, and once again by allocating separate chapters to these subjects he can be seen as excluding them from his explicit account of geography.

In complete contrast to the works of Strabo and Pliny is the Geography of Claudius Ptolemaeus (c. AD 90-168) known to subsequent generations as Ptolemy (Ptolemy, 1966; Rylands, 1893). Above all else, Ptolemy was an astronomer, who worked at Alexandria, and before turning his attention to geography he had written a compendium of astronomy, subsequently known by the Arabic name *Almagest*. In this, he assumed that the globe is at rest in the center of the universe, and among other information he provided a detailed table of thirty-three latitudes up to the Arctic Circle (Thomson, 1948; Dreyer, 1953). In the opening chapter of his geography, Ptolemy defines geography as an imitative delineation of that part of the earth comprehended within our knowledge as a whole, with its parts roughly (lit. Generally) appended' (Rylands, 1893: 18), and he is at pains to contrast this with chorography considers only the part. Moreover, for Ptolemy the mathematical sciences are central to geography, whereas chorography that is merely concerned with topographic description has no need of them. Having explained how to construct a globe, together with its parallels and meridians, Ptolemy then provided details of two projections, one conical and the other globular, which can be used to project the known world onto a plane surface.

Once the theoretical outline was completed, he then

devoted the bulk of his geography to a division of the continents into regions and to a listing of the latitudes and longitudes of all of their major features. This vast compendium provided an atlas which was to form the basic account of the European world until the Portuguese and Spanish explorations of the 15th and 16th centuries. Unfortunately, despite its apparent precision, it was based on fundamental inaccuracies. As Bunbury (1879;) has commented, the means at his command did not enable him to carry his ideas into execution; the substance did not correspond to the form; and the specious edifice that he reared served by its external symmetry to conceal the imperfect character of its foundations and the rottenness of its materials'. A fundamental inaccuracy entered his work because Ptolemy relied for his calculations of longitude on the maximum circumference of the earth as calculated by Posidonius at 180,000 stades. Ptolemy's adoption of this figure is somewhat surprising because Eratosthenes had earlier calculated it much more accurately at 252,000 stades. There are fundamental problems in equating the Roman stade (stadion) into modern measurements, but that used by Ptolemy is usually (Thomson, 1948: 161; Dreyer, 1953: 175; Holt-Jensen, 1988: 12) seen as being equivalent to 157.5 metres. If this figure is used, the difference between the figures of Posidonius and Ptolemy amounts to some 11,340 kilometers. Ptolemy's measurements thus underestimated the true circumference of the earth by some 11,729 kilometers, and this was one of the reasons that medieval explorers using Ptolemy's calculations vastly underestimated the size of the globe.

Activity

Write down the major contributions of Strabo, Ptolemy and Pliny to the development of geographic thought.

Summary (The Greek and Roman concept of geography).

Most commentators on the Greek and Roman contributions to geography (Bunbury, 1879; Thomson, 1948) contrast the innovative, dynamic and original contributions of the Greeks with the utilitarian replicative and stagnant geographies of the Romans. Gould (1985: 13) has thus commented that 'Unfortunately, the marvelous intellectual and artistic efflorescence of the classical and Alexandrian Greek worlds could not last. The Roman spirit was of a different order, and fundamentally inimical to the Greek tradition of probing enquiry, the sort of inquiry that was prepared to go wherever the questions might lead. It is from the geographies of Strabo and Ptolemy, both Greeks writing at the height of the Roman Empire, that the remnants of classical geography were to survive through the early medieval period. Their geographies represented the marked contrast that existed within the discipline at that time. Strabo's geography was of the type decried by Ptolemy as a mere chorography. However, the basic questions tackled by geographers ever since were firmly considered by the geographers of the Greek and Roman world.

Three central issues lay at the heart of classical geography. First there was a concern with topographical description, or what Ptolemy called chorography. This involved the detailed

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description of places, accounting for the genesis of human occupancy therein, and thus including an element of historical understanding. Second, and related to this was

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a mathematical and astronomical concern with measurement, represented in its purest form, for example, by Ptolemy's Geography. This involved not only the measurement of the earth's dimensions, but also considered the world's position in the universe, and its relationship with the stars and planets. Third, though, geographical considerations were also philosophical ones, concerning the origins of nature. In particular, geographers sought to understand the place of humanity within the natural world.

All of these considerations were tempered and conditioned, though, by an underlying appraisal of the role that geography played within society. This was made explicit in Strabo's account of the utility of geography, but the use of knowledge about different places and peoples was intimately tied up with the exercise of political control throughout the Greek and Roman worlds. This was well illustrated in the close relationships between geography and conquest, both in the eastern campaigns of Alexander, for example, which took him as far east as the Indus, were in part enabled by earlier geographical writings, but also led to a considerable subsequent expansion in geographical knowledge as expressed in the writings of Dicaearchus and Eratosthenes.

Definition of Key words and concepts

Periodus: This is a term/book meaning description of the earth and was the first major description of the earth written by Hecataeus and divided into two books entitled Europe and Asia. Republic and Timaeus: These are the two great works by Plato in which he describes the earth as a sphere. Geographica: This is a three-volume treatise published by Eratosthenes (the parent of scientific geography) in which he gives topographical and

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astronomical accounts of the earth.

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Strabo: A geographer of the ancient times who has given detailed topographical description of the known world.

Ptolemy: An astronomer who contributed a lot to geographic thought during the ancient times.

Almagest: Arabic name given to a compendium of astronomy written by Ptolemy.

Stade (stadion): Measurements used by ancient Greek and Romans to measure the circumference of the earth. One stade (stadion) is equivalent to 157.5 metres.

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LECTURES 7,8, &9

GEOGRAPHY IN THE ANCIENT WORLD: CHINESE AND ISLAMIC GEOGRAPHY.

Introduction

In the previous lessons, we have looked at Greek and Roman geography during the ancient times. In the early Greek writing we have noted three geographical traditions namely topographical, mathematical and astronomical; and a theological tradition. We have also looked at the contributions of some Greek and Roman scholars to the development of geographic thought. In particular we have examined the contributions of Herodotus, Plato, Eratosthenes, Aristotle, Strabo, Pliny, Ptolemy and Pythagoras.

In these lectures, I will try to give a description of geography in the Chinese and Islamic world. I will explain the five main types of Chinese geography and also look at the Islamic contribution to geographical understanding.

Objectives

By the end of these lectures you should be able to:

- a) Name the five main types of Chinese geography
- b) Pinpoint the most important contributors to Chinese astronomy and cartography.
- c) Explain the fundamental factors influencing Islamic geographical writing.
- d) Name the early Islamic geographers and describe their contributions to geographical understanding.

Chinese geography: a separate tradition.

With the collapse of the Western Roman Empire in 476, European geography entered a period of dark stagnation. A few copies of Greek and Roman geographical texts survived, but the Germanic tribes that overran the vestiges of the Roman world had little use for such works. Indeed, many had already been lost, particularly at Alexandria, where the fire of 47 BC had destroyed some 400,000 manuscripts in the great library, and the disturbances of AD 391 had led to the loss of perhaps 300,000 more works in the Temple of Serapis (Cornell and Mathews, 1982; Gould, 1985). Parallel to, but totally separate from, the Greek and Roman world a completely different culture of science had meanwhile evolved to the east in China. It was here that the subsequent focus global intellectual and scientific activity was to be encountered, particularly under the Tang (AD 618-907) and Southern Sung (1127-1279) dynasties, the latter of which was described so magnificently by the Venetian Marco Polo. Once again, the emergence of a tradition of geographical writing at this time can be seen in part to have been influenced by military conquests and the need for the emperors to have a sound knowledge of their lands in order to retain their positions of power. Moreover, the development of new surveying and cartographic skills enabled the Chinese to produce maps of a quality far surpassing anything being produced in medieval Europe.

The earliest Chinese geographical document is reputed to be the Yü Kung (Tribute of Yü) chapter within the *Shu Ching* (Historical Classic) dating from the 5th century BC. This provides an inventory of the Chou Empire mainly in terms of its physical geography, and

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lists 'the traditional nine provinces, their kinds of soils, their characteristic

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products, and the waterways running through them' (Needham and Wang Ling, 1970: 500).

Other ancient travellers' guides, such as the *Shan Hai Ching* much of which dates from the 4th century BC, can also be considered to be geographical, but most include mythological and magical elements together with details of semi-human races and peoples. Needham and Wang Ling (1970) suggest that there were five main types of Chinese geography:

1. Anthropological geographies, known as *Chih Kung Thu* (Illustrations of the Tribute-Bearing Peoples), dating from the mid-6th century AD.
2. Descriptions of the folk customs of the countries to the south of China (*Feng Thu Chi*) and descriptions of unfamiliar regions (*I Wu Chih*) both dating from the 2nd century AD;
3. Hydrographic books and coastal descriptions, such as the *Shui Ching* (Waterways Classic);
4. Local topographies or gazetteers, such as the *Hua Yang Kuo Chih* (Historical Geography of Szechuan) which were mostly written from the 4th century AD onwards;
5. Geographical encyclopaedias compiled from the Chin dynasty (3rd and 4th centuries AD) onwards, in a style similar to that of Strabo.

Question

Do these kinds of geographies still exist today?
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Chinese geography was also closely tied to astronomy and cartography. Astronomy played a central role in Chinese science because of its religious concern with cosmic

unity, and its links with astrology. Moreover, knowledge of astronomy and the compilation of agrarian calendars were also a way in which the state could control the productive capacity of the population. Traditionally, there was an ancient belief in China that the heavens were round and the Earth Square, but by the 2nd century AD three main schools of cosmology and astronomy had emerged (Needham and Wang Ling, 1970): the Kai Thien theory which envisaged the heavens as a hemisphere covering the earth which was shaped like an upturned bowl; the Hun Thien school, which corresponded with the Greek view of heavenly spheres revolving around the globe; and the Hsuan Yeh teaching which envisaged an infinite space in which the heavenly bodies floated freely about.

Among the most important contributors to Chinese astronomy and cartography were Chang Heng (AD 78-139) and Phei Hsiu (AD 224-271). Although maps from as early as the 3rd century BC are recorded in China, it was these two scholars who first developed a scientific method of cartographic work in the ensuing centuries was then brought together by Chu Ssu-Pen (1273-1337), who used it to summarize the wealth of new information which had become available as a result of the Mongol unification of Asia. His map of China prepared between 1311 and 1320 is a remarkable achievement, and remained a basic work of reference for over two centuries. Although Chu Ssu-Pên was cautious about depicting lands distant from china, it is evident that he had a level of knowledge surpassing anything current in Europe at that time. He thus recognized that Africa was a southward pointing triangle, whereas on contemporary European and Arabic maps it was always represented as pointing eastwards (Needham and Wang Ling, 1970).

Activity

Select any two Chinese geographers and give an account of their contributions to Chinese astronomy and cartography.

The Islamic contribution to geographical understanding.

Situated between Europe and China, the explosion of energy that accompanied the emergence of Islamic power in the 7th and 8th centuries AD also saw the creation of another great tradition of geographical scholarship. Although the first waves of Islamic conquest led to the destruction of many ancient archives, such as those that remained at Alexandria, the subsequent emergence of centres of scholarship at the cities of Cairo, Damascus, Baghdad and Granada, for example, formed the focus for much geographical writing. Not only were surviving Greek and Roman texts translated into Arabic, but with their trading contacts to the east the Arabs were also open to cultural and scientific influences from China.

One fundamental factor influencing Islamic geographical writing was the religious requirement for the faithful to make the pilgrimage to the holy places in the vicinity of Mecca at least once in their lives (Robinson, 1982). This gave rise to numerous travel guides, which provided pilgrims with descriptions both of the journeys and of the holy places of Arabia themselves. Moreover, this requirement to travel brought scholars from all over the Arab world into contact with each other, and provided a forum for considerable intellectual debate. Many of the works of early Islamic geographers, such as Ibn Khurrahbih (ft.

850), al 7aqubi (ft. 900) and Ibn Hawkal (ft. 953) (Baker, 1937),

had been based on compilations of previous writes, but with al-Muqaddasi (c. 945-c. 988) a new emphasis was introduced. He had traveled widely himself and is generally acknowledged to have been among the first to insist that all of the information he presented should be based on his own experiences (Scholten, 1980; Holt-jensen, 1988). Few of these early Arabic geographers however had any influences on Christian Europe.

In contrast, al-Idrisi (Abu 'Abd Allah Muhammad B. Muhammad B. Abd Allah B. Idris al-Ali Bi-Amr Allah, called Al-Sharif Al-Idrisi) (1099-1180) (Encyclopedia of Islam, 1971), who completed his descriptive geography entitled *Kitab Nuzhat al-mushtak fi ' Khtirak al-afak* in 1154, was to have much greater influence because of his connections with the court of Roger II of Sicily. This text had been written to accompany a large silver planisphere which he had previously constructed for Roger II, and in working for one of the most powerful Christian kings of the time he was also able to gain information on Europe that had been unavailable to previous Islamic geographers. In his preface, al-Idrisi outlines his task as being to describe the towns and territories shown on the planisphere, the nature of the agriculture and settlements, and the extent of the seas, mountains, rivers and plains (Jaubert, 1975; xxi). Moreover, he goes on to include details of the main crops, the arts and crafts, the import and export trade, and the customs, curiosities and religions of the people living within each of the seven climates into which he divides the world. This is a remarkable undertaking, and there are few other surviving accounts of Europe in the first half of the 12th Century to match it. The best known Arabic travel writer was Ibn Battuta (Shams al-Din Abu 'Abd Allah Muhammad B. 'Yusuf al-Lawati al-Tadji) who was born at

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Tangier in 1304 and who died in Morocco in

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1368-69 or 1377 (Encyclopedia of Islam, 1971). His *Tuhfat al-nuzzar fi ghard ib al-amsar wa-adja ib al-asfar* was completed in 1357 and was of major significance for its descriptions of India, Anatolia and West Africa. Moreover, during the wide travels, Ibn Battuta was at one time appointed by the ruler of Delhi as ambassador to the court of China, and as a result he also visited the Maldivian Islands and Ceylon on his journey to the east (Sykes, 1934; Baker, 1937).

Ibn Khaldun (Wali Al-Din 'Abd al-Rahman B. Muhammand B. Muhammad B. Abi Baki Muhammad B. Al-Hasan) (1332-1406), a contemporary of Ibn Battuta, is also sometime considered to have been a geographer (Holt-Jensen, 1988), but he is best regarded as a historian, a philosopher of history, and a proto-sociologist' (Morgan, 1988; 202). Despite this, his great cyclical view of world history, the *Mukaddimah*, provides a good overview of the state of Arabic thinking concerning geography in the 14th century (Ibn Khaldun, 1967). Although the main focus of Ibn Khaldun's writing concerned the processes of state formation and decline, he developed his ideas through a consideration of the physical environment which he saw as forcing people to live together in social and political groups. Central to his argument was the view that states develop through a natural sequence of growth, maturity, decline and fall, because group solidarity is inevitably eroded by the process of civilization. Ibn Khaldun's geographical account is mainly included in Chapter 1, which explicitly acknowledges his debt to Ptolemy and al Idrisi. Indeed his map and subsequent topographical description of different parts of the world is based almost entirely on that of al-Idrisi. Within his second preparatory discussion he describes the earth as

having a spherical shape and being enveloped by water. He again

uses the sevenfold division of the cultivated area of the world, noting that the northernmost and southernmost zones are least populous and civilized. For Ibn Khaldun (1967; 1. 104) civilization has its seat between the third and the sixth zones. In developing this argument he explicitly accounts for the distribution of civilization by the harshness of the climatic regime and physical environment at the two extremes; in the south it is too hot, and in the north too cold. Moreover, elsewhere in his fourth preparatory discussion he also argues that climate has a significant direct influence on human character: people living in hotter coastal and southern regions are described as sad and gloomy (Ibn Khaldun, 1967: 1. 174). Such ideas were to have a lasting influence on geographical thought, and survived through their incorporation into environmental determinism well into the 20th century.

As well as topographical accounts and cartographic descriptions of the world Islamic scholars also added considerably to the body of astronomical writing, although there is much debate concerning their original contributions in this field (Mieli, 1938; Dreyer, 1953; Needham and Wang Ling, 1970). Again, religion played a significant role in determining the importance given to astronomy, both because of the determining the importance given to astronomy, both because of the need to determine lunar movements for the calendar, and also for the identification of the direction to which the faithful should turn for prayer. It seems that Arab astronomy was based both on ideas introduced from India during the 8th century and also on the rediscovery and translation of earlier Greek texts. In particular, Islamic astronomers sought to revise the works of Ptolemy and Aristotle, but this they were largely unable to do because they failed to recognize that the

earth was not at rest. Their main astronomical achievements thus appear to have been in the general development of a tradition of Greek scholarship which had disappeared in Europe in the face of Christian dogmatism, and also in the construction of observatories such as those at Damascus, Baghdad and Cairo.

The Chinese and Islamic contributions to geography in the first millennium AD represented fundamental advances on anything happening in Christian Europe. The Chinese had developed a culture of science in which knowledge was used extensively in the support of the Emperor's wishes. Great advances were made in cartographic representations of the Empire, and topographical accounts provided surveys both of the human and physical characteristics of the regions under the Emperor's way. Islamic science was, in contrast, closely influenced by religious requirements. This was particularly so with respect to astronomy, but topographic accounts and cartographic skills were also largely related to the uniting of the Islamic world, and an interpretation of divine instruction.

Activity

Name any three Islamic geographers and describe their contributions to geographical understanding

Summary

In these lessons we have looked at the Chinese and Islamic geography. We have noted the five main types of Chinese geography namely: Anthropological geographies, descriptive geographies of the folk customs, Hydrographic and coastal geographies, local

geographies or gazetteers and geographical encyclopaedias. We have also noted two important contributors to Chinese astronomy and cartography namely Chang Heng and Phei Hsiu. We have further examined the contributions of Islamic geographers to geographical understanding noting the contributions of Al-Idrisi, Ibn-Battuta, and Ibn Khaldun.

Definition of Key words and concepts.

Cartography: the production and study of maps and charts.

Chang Heng and Phei Hsiu; These were Chinese scholars who first developed a scientific method of cartography based on a rectangular co-ordinate system.

Ibn Battuta and Ibn Khaldun: These were Islamic scholars and travellers who gave detailed topographical accounts and cartographical descriptions of the world.

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LECTURES 10,11,&12.

**GEOGRAPHY IN THE CHRISTIAN WORLD: THE RESURGENCE
OF EUROPEAN GEOGRAPHY.**

Introduction.

In the previous lectures, we have given a description of geography in the Chinese and Islamic world. We have looked at the five main types of Chinese geography and also at the Islamic contribution to geographical understanding, in particular we have noted the contribution of three Islamic scholars and travellers namely Ibn Battuta, Ibn Khaldun and Al-Idrisi.

In these lectures, I am going to look at geography in the Christian world. In particular, I will look at the resurgence of European geography during the middle ages.

Objectives

By the end of these lectures, you should be able to:-

- a) Identify the influence of Christianity on medieval cosmology.
- b) Describe the salient characteristics of the early medieval Christian world maps.
- c) Explain the expansion of European power and its influence on Geography.
- d) Give an overview of Geography at the dawn of the 17th century.

Question

What factors led to the collapse of the Roman empire.

Following the collapse of the Roman Empire, the dominance of Christianity in Europe led to a rejection of much of the earlier scientific scholarship of the Greek world. If Roman culture had led to a stagnation in the inquiring, critical spirit of the Greek world, then many Christians sought completely to destroy it. In Dreyer's (1953: 207) words, A narrow-minded literal interpretation of every syllable in the scriptures was insisted on by the leaders of the Church, and anything which could not be reconciled therewith was rejected with horror and scorn, Although the earliest Christians do not appear to have been antipathetic towards science, the elevation of Christianity to the position of the state religion of Rome under Constantine in the 4th century led to the propagation of doctrines that were inimical to critical scientific enquiry. In particular Lactantius (ft. 4th century) in his *Divinarum institutionum*, declare science to be both foolish and false (Kimble, 1938: 14), and sought to ridicule the belief that the world was spherical (Dreyer, 1953).

However, not all Christian rejected earlier scholarship, and within the growing number of monasteries that began to flourish throughout the Christina world, surviving ancient manuscripts wee studied and transcribed. Thus the anonymous Geographer of Ravenna, writing in the mid-7th century, although a devour Christian, made extensive use of the works of pagan scholars, and the English monk Bede writing at the beginning of the 8th century copied large extracts from Pliny's work in his *De natural rerum* and his *De*

temporum ratione. Bede, for example, appears to have accepted the spherical form of the

earth, and the annual motion of the sun and planets, even though he also acknowledged the Christina belief that there was water surrounding the heavens (Dreyer, 1953).

The dominance of Christianity also had significant ramifications for medieval cosmology. Of most importance was the Judaeo-Christian teleological belief in God's creation and care for the universe as expressed in Genesis (Glacken, 1967; Doughtly, 1981). This view consisted of four central elements. First, the world was created out of chaos by God (Genesis 1). Thus, in Glacken's (1967; 153) words the creation is a continuing process requiring the constant care, activity, and solicitude of God. Second, humanity, made in the image of God (Genesis 1:27), is set apart from nature. There is therefore a clear divide between the realm of nature and the realm of people. Third, the products of nature exist for humanity, and it is the task of people to cultivate and maintain them; people are here seen as caretakers of the environment (Genesis 2:15). Fourthly, as a result of the Fall, when Adam and Eve disobeyed God, the idyllic relationship between people and the environment is broken and thereafter humanity has to toil in a cursed environment (Genesis 3:17). Once more, in Glacken's (1967:153) words.

The story of the fall became important to the Christian idea of nature because it is the source of the belief, widely held through the seventeenth century, that the fall of man has caused disorder in nature and a decline in its powers, an idea clearly distinct from the classical idea of a senescence in nature based on the organic analogy.

These four elements were to play a fundamental role in shaping the subsequent development of geographical ideas concerning the relationship between people and the environment.

From the 12th century onwards, the resurgence of European scholarship and science was underlain by three important processes: the reconquest of Iberia from the Moors beginning in the late 11th century and leading to the incorporation of elements of Islamic science into Christian European understanding the Portuguese and Spanish voyages of discovery for the 15th century onwards which opened up whole new areas of the world to European scientific examination; and the expansion of the Ottoman Empire and the fall of Constantinople in 1453 which caused a considerable migration of eastern scholars and knowledge in Europe.

Activity

Name the three important processes that led to the resurgence of European scholarship and science from the 12th Century onwards.

A world of discovery: cartography and Exploration.

By the beginning of the 15th century, with the establishment of the House of Avis as the ruling dynasty of Portugal, European horizons were dramatically altered by the so-called Age of discovery. Following the economically draining wars with Castile, the Portuguese King Joao I, and his sons Dom Duarte, Dom Pedro and Dom Henrique, turned to overseas exploration as a way of countering many of the problems that beset the country, In particular, any economic gains would provide urgently needed income, long voyages would

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keep a restless nobility active outside the country, and foreign adventure would

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distract attention from internal social problems (Bell, 1974; Diffie and Winus, 1977; Unwin, 1987). From the capture of Ceuta in North Africa in 1415, the Portuguese gradually ventured further around the coast of Africa, until Bartolomeu Dias reached the cape of Good Hope in 1488. Many of these early voyages were sponsored by Dom Henrique (1394-1460), known in English as Prince Henry the Navigator, who established a school of navigation and cartography on the Sagres Peninsula in the far south-west of Portugal. Dom Henrique's central purpose appears to have been to increase his own financial resources, and although he did play an important role in bringing together sailors, navigators and adventurers to undertake these voyages of discovery he was not alone, and many other voyages were sponsored by the King and nobility. This Portuguese expansion was enabled by two main factors: the culmination of sailing experience derived from previous commercial and fishing voyages in the north Atlantic (Diffie, 1960), and the acquisition of Islamic geographical knowledge.

Towards the end of the 15th century the pace of discovery changed dramatically. In 1492 the Genoese Christopher Columbus, in the service of the Castilian Crown reached San Salvador, Cuba and Haiti (Sykes, 1934; Baker, 1937). India was attained during Vasco da Gama's voyage from 1497 to 1499 (Ravenstein, 1998), and Carbal crossed the Atlantic to Brazil in 1500. Portuguese ships sailed to China in 1513, and in 1519 Fernando de Magalhaes set sail on his circumnavigation of the globe, which was at last firmly to shatter any remaining illusion that the earth was flat (Diffie and Winus, 1977). These voyages were both enabled by the rediscovery of classical geographical and cartographic works,

but they also rapidly led to a fundamental revision of methods of map making (Crone, 1968).

In 1295 Maximus Planudes (c. 1260-13010), a monk at the Chora monastery in Constantinople, had eventually rediscovered a manuscript of Ptolemy's Geography (Dilke, 1985). This manuscript had no maps with it, and Planudes therefore began compiling ones to accompany the text. However, during the 14th century knowledge of Ptolemy's work remained in constantinople, and it was only in 1406 in Florence that the Tuscan Jacop d'Angelo da Scarperia translated the work into Latin (Dilke, 1985). Manuscript versions of the maps were soon prepared, and from the 1470s printed editions of the text and maps together were produced at Bologna, Rome and Ulm. Revisions, however, were soon necessary. Ptolemy had suggested that the Indian Ocean was enclosed by a piece of land running from Africa to the Maly Peninsula, and it was thus the voyages of Dias and da Gama that definitively proved this to be wrong. Moreover, Columbus recalculated the size of the globe to include Marco Plo's eastern discoveries, and although he reached America rather than Asia, it was these calculations based in part on those of Ptolemy that had given him the incentive to sail westwards.

Ptolemy was not, though, the only influence on medieval cartography, and long before the rediscovery of his works European cartographers were grappling with ways of depicting the world in which they lived (Harvy, 1980). Early medieval Christina world maps were often very simple representations, without any indication of latitude or longitude, designed

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as much to depict theological as topographical truths. The most

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common, known as T-O maps, depict the three known continents separated by a T, in the midst of the surrounding ocean O. Similar to such maps, but less rigidly divided, is the Hereford Map designed by Richard of Haldingham in the late 13th century. As with most Christian maps, Jerusalem was in the center, and Dilke (1985) suggests that the prominence given to places of pilgrimage and the Alpine passes 'suggest that one object of the Hereford Map was to act as a help to intending pilgrims'. Other maps, such as the mid-13th century maps of Britain and the Holy land by Mathew Paris, appear likewise to have been designed essentially as itineraries (Harvey, 1980).

Itemize the contributions of the following explorers to geographic thought.

- Bartolomeu Dias
- Christopher Columbus
- Vasco da Gama
- Fernando de Magalhaes
- Marco Polo

Activity

Geography and the expansion of European power

The rediscovery of classical cartography was central to the expansion of European political and economic power in the 15th and 16th centuries. However, although Ptolemy's great work was called *a Geography*, there were few medieval or renaissance explorers who called themselves geographers. Likewise, although a strong tradition of topographic writing had developed in the wake of work such as the late 12th century Topographical Hibernica (Topography of Ireland) written by Giraldus cambrensis (1951), these made few claims to be works of geography. Thus, although the astronomical, cartographic, cosmological and

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topographic traditions of classical Greek and Roman geography re-

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emerged in the medieval period, there was no widely accepted medieval definition of geography, and much so called medieval geography has merely been given this title by subsequent commentators (Wright, 1925; Taylor, 1930). The nearest word encompassing the essence of later geography was cosmography, and during the 15th and 16th centuries a number of treatises bearing this title were published, often modelled on the works of Strabo and Ptolemy. Among the most famous of these are the *Cosmographiae introductio* of Martin Waldseemüller, the *Cosmographicus liber* of Petrus Apianus and the *Cosmographia universalis* of Sebastian Münster (Dickinson, 1969).

The beginning of the 16th century saw a fundamental departure from classical understanding of cosmology and astronomy. Around 1530 Nicolas Copernicus (1473-1543) eventually finished the manuscript of his work *De revolutionibus*, the basic ideas for which had probably first been thought out soon after 1506 (Dreyer, 1953; Babicz, Buttner and Nobis, 1982). In this, Copernicus put forward an entirely new theory of planetary movement, which countered the long accepted arguments of Aristotle and Ptolemy by demonstrating that the earth was not the center of the universe, but instead revolved around the sun together with the other planets. This had important ramifications for the links between geography and astronomy, because while the earth was still considered to be the center of the universe it was reasonable to include discussions about that universe in geographical writing about the earth. Astronomy could thus logically be considered an extension of geography, however, following the acceptance of a heliocentric view of the universe, astronomy threw off the shackles of its earth-bound past, and simultaneously

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broke its links with geography.

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The vast expansion of cartographic and topographic knowledge in the 16th century opened up whole new avenues of commercial and political action. With the demise of the Iberian powers, and the emergence of the Dutch and English as the most formidable European states by the beginning of the 17th century, the heart of the impetus for further discovery and exploitation shifted northwards. Thus, in referring to the emergence of geographical ideas in 16th century England, Tayler (1930;v) has noted that, Elizabeth's day saw the map and the globe as the necessary furniture of the closet of scholar, merchant, noble and adventurer alike, and the dreams of Empire were formulated with found expression in Drake's achievement and Humfrey Gilbert's splendid failure'. Maps, globes, topographic accounts, and the few explicit geographies that were written, such as the geographia (1540-41) of Roger Barlow, were designed with one purpose mainly in mind: that of enabling English ships to sail the furthest oceans and bring back the products of China, India and the Americas for commercial profit.

These tendencies found tier fulfilment in the writings of Richard Hakluyt the Younger (1552-1616) and the Reverend Samuel Purchas (1577-1626). Hakluyt lectured on cosmography at Oxford, and had benefited greatly from his contacts with other cosmographers, particularly through the practical work of the Portuguese and the theoretical schemes of the Flemish school (Taylor, 1934). However, he described his great work, the principal navigations, voyages, traffiques and discoveries of the English nation (1589) as a history (Hakluyt, 1903; xxiii, xxx), and in its preface he confines the term

'geographical' to the depiction of places on a map (Hakluyt (1903: xxx). Indeed, in

the preface to the second edition of 1598 Hakluyt (1903; xxxix) makes this distinction more explicit by referring to 'geographie and Chronologie (which I may call the Sunne and the Moone, the right eye and the left of all history). Here geography is used to refer each incident or voyages of English people abroad and to weld them into a memorial of the achievements of the English nation. The first edition confined itself to English voyages of discovery, colonization and commerce, but in the three volume second edition of 1598-1600 he also included material on naval engagements, particularly the defeat of the Spanish Armada, and some details of voyages by foreigners where they provided additional information on parts of the world not elucidated by English authorities. Hakluyt's intentions in writing his *Principal navigations* were to provide a eulogistic account of the achievements of the English and to preserve documents which would otherwise have been buried in perpetual oblivion (Hakluyt, 1903:xxxii). However, underlying these he was eager to promote continued English overseas expansion, particularly in the newly established colonies of Virginia, and in his later life he developed this practical intent, serving as an adviser on Virginia and East India matters (Taylor, 1934:33).

Hakluyt's role as a collector of travel accounts was taken up by Samuel Purchas, who had been educated in divinity at St. John's College, Cambridge. Remarkably, unlike Hakluyt, he never traveled outside England, but his publications brought together a wealth of material concerning the world, as it was known in the 17th century. His first book, *Purchas his pilgrimage or relations of the world and the religions observed in all ages and place discovered*, from the creation to this present was published in 1613, and sought to

combine his theological and geographical interests by providing a synthesis of the different religious practices of the world's peoples. This work was immediately popular, and was even read by King James (Taylor, 1934). As a result he met a number of influential people, such as Sir Walter Raleigh, who furnished him with yet further manuscripts and books. In particular, Hakluyt gave him access to his own collection of manuscripts, and these were important source for his later work, Hakluyt's posthumous or Purchas his pilgrimes containing a history of the world in sea voyages and land travels by Englishmen and others, which appeared in two volumes in 1625, nine years after Hakluyt's death. While these volumes bring together a mass of secondary sources concerning voyages and travels, they do not claim strictly speaking to be geography. As in his earlier work, Purchas was seeking to incorporate geography works into a historical account. Like Hakluyt, his scholarship was imbued with a practical concern with colonial expansion, and by bringing together the results of explorations and discoveries of the New World he was seeking to inspire further English settlement overseas.

Activity

Write a brief essay on the contributions of Hakluyt to the development of geographic thought scholarship
--

Geography at the dawn of the 17th century.

In 1600 geography was a term of many meanings, conjuring up a range of different images to those who used it. At its simplest it referred to topographical writings about particular places, and thus included descriptions of distant lands, coastlines and harbors. In effect this

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type of geography was none other than Ptolemy's chorography, whereas

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Ptolemy's concept of geography had become subsumed within cosmography, the science of the globe. However, such descriptions required precise elucidation, and since classical antiquity this had meant that they were frequently associated with mathematical and astronomical calculations. Cartography thus formed a central part of geography, providing the detailed basis upon which voyages of discovery could be made and subsequent descriptions of new lands could be written. These links with mathematics and astronomy also brought the subject into a close alliance with astrology, which sought to explain human and environmental phenomena through reference to astronomical phenomena. Indeed Livingstone (1990a, b) has argued that many 16th and 17th century writers about geography such as William Cunningham and John Dee (Matley, 1986) were deeply implicated in magical practices (see also Sack, 1976). This view of geography owed much both to the mystical ideas of Pythagoras and to the writings of subsequent generations of Islamic scholars.

Above all geography had retained the practical importance attributed to it by Strabo. It was essential for successful merchants and statesmen alike, providing the basis upon which European nations came to dominate the globe. The rediscovery of Greek and Roman geographical works, enhanced by later Arab writers, had opened up new visions of the world to the Portuguese and Spanish explorers of the 15th and 16th centuries. In turn, their discoveries overthrew long established traditions and images, and provoked a range of new questions for the emergent sciences of the 17th and 18th centuries to answer. In short, as Livingstone (1990a:8) has argued; First hand encounter with the world the very stuff of

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geography brought an immense cognitive and cultural challenge to tradition. But

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geography, as yet, remained poorly defined. There was not the institutional dominated concern with the clear delimitation of disciplinary boundaries that was to be encountered in the 19th century, and anyone who wrote about the earth could claim to be a geographer. It was not until the 17th and 18th centuries that a formal geography to compare with that of Strabo or Ptolemy began to emerge.

Summary

In these lectures, we have tried to look at the resurgence of European geography. We have seen how the dominance of Christianity led to significant ramifications for medieval cosmology. We have also noted how the age of discovery altered geographical thinking. We have further examined the contributions of early explorers to geographic thought. Finally we have briefly looked at Geography at the dawn of the 17th century and noted that above all geography had retained the practical importance attributed to it by strabo.

Definition of key words and concepts.

- De natura rerum and De temporum ratiore:- These were great European works written by the English monk Bede who copied large extracts from pliny's work.
- Teleology: This is in literal terms, the study of purposes, goals or ends.
- Hakluyt: A cosmographer who lectured at Oxford and made significant contributions to the development of geographic thought in the Christian world.
- Purchas: A close counterpart of Hakluyt who incorporated geographical works into a historical account in the Christian World.

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LECTURES 13,14, &15

GEOGRAPHY DURING THE RENAISSANCE PERIOD (15TH-17TH. CENTURY)

Introduction

In the previous lessons, we have looked at the development of Geography during the middle ages (5th-15th century). We have noted that the middle ages were a dark period in the development of Geography in Europe. We have further noted that, at best, scholars during this period made accurate and sterile copies of the works of the ancients, rejecting anything that did not conform to the dogmas of the church. Such intellectual environment stifled any development of critical scientific analysis. We have also noted that concepts of the world, which had been developed in the ancient times, were reshaped to conform to the teachings of the church. We have finally noted that during the middle ages, the earth was regarded as a flat disc with Jerusalem at its center.

In these lessons, we are going to look at geographical developments that took place in the Renaissance period (15th- 17th century). We shall briefly look at geography during the age of exploration and discovery. We shall also look at the contributions of two eminent

geographers, namely *Bernhard Varenius* and *Immanuel Kant*, to the development of Geography during this period.

Objectives.

At the end of these lessons, you should be able to:

- (a) Describe the development of Geography during the age of exploration and discovery and
- (b) Discuss the contributions of *Bernhard Varenius* and *Immanuel Kant* to the development of Geography.

Geography during the age of Exploration and Discovery.

? Write down names of great travelers and explorers that you know of.

During the age of exploration and discovery Arab traders traveled widely gathering information that was used to improve *Ptolemy's* original map. The best known of these Arab travelers was *Ibn Battuta* (1304-1368) who traveled as far east as Peking in China and sailed far south of the equator along the east coast of Africa. The travels by *Ibn Battuta* proved Aristotle wrong in believing that it was too hot for human habitation in what the greeks had called the 'torrid zone'. By the 12th century, *Al Idrisi* had shown that the Greek division of the world into five climatic zones (two cold, two temperate and one torrid) did not correspond to reality and had suggested a more sophisticated world climatic system.

Another great Islamic geographer of the Middle Ages was *Ibn Khaldun* (1332-1406). He established the foundation for historical geography in some of his writings that analyzed

the rise and fall of empires. The works of *Idrisi* and *Khaldun* were not translated into Latin or any other west European language until the 19th Century and consequently the Europeans were unable to make use of the observations of these Arab scholars.

The journeys undertaken by the Europeans during the middle ages made little significant contribution to the development of geographical knowledge. Around 1000 A.D. the *Norsemen* sailed across the North Atlantic to Greenland and North America but the findings of these adventures were only passed on by word of mouth and written down long afterwards in Iceland.

Exploration and discovery also flourished in China, Europe and India were 'discovered' by Chinese travelers long before the European reached the Orient. Between 2nd century B.C and 15th century -A.D. the Chinese culture was the most efficient in the world in applying knowledge of nature to useful purposes. At this time geographical studies were more advanced than in Europe and the Chinese used coordinates and triangulation to produce beautiful maps of China and neighboring countries. When, however, the Italian Marco Polo (1255-1325) wrote an account of his travels to China describing the high travel of Chinese learning, his book was dismissed as fictitious adventure.

Activity

Write a brief essay on the contributions of the early travelers and explorers to the development of geographic thought.

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Geography During the Renaissance period.

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The Renaissance brought about a renewed interest in the geographical knowledge of ancient times. Copies of *Ptolemy's* geography that had been preserved at Istanbul were discovered and brought to Venice where they were translated into Latin. Christopher Columbus and other expeditionary leaders relied on *Ptolemy's* calculations, but some of their discoveries dis-approved *Ptolemy's* calculations of longitude and changed the world picture he had established. During the Renaissance period there were new developments in cartography. New map projections were invented, the first globes were made and New World maps were published.

During the renaissance period there was also a revival in another branch of ancient geography, that of topographic description. Large numbers of accounts of voyages provided raw materials for encyclopaedic works on the geography of the world.

Activity

Briefly enumerate the major geographical developments that took place during the renaissance period.
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Contributions of *Bernhard Varenius* and *Immanuel Kant* to the Development of Geography.

The importance of *Bernhard Varenius* and *Immanuel Kant* to Geography is that they were among the pioneers of the subject. The ideas of these two scholars have been of great influence in shaping and developing the subject into one of the sciences. *Varenius* and *Kant* devoted great attention in organizing and developing geography. *Varenius* gave geography

its theoretical base while *kant* gave it its philosophical foundation.

***Bernhard Varenius*: (1622-1650)**

Varenius was born in Eastern Germany in 1622 and went to the Netherlands where he studied medicine and became interested in Geography. Basing his work on critical use of reports of the first Dutch pioneers in Japan, *Varenius* wrote a book on the Geography of Japan. But it is on the basis of *varenius'* book *Geographia Generalis* (General Geography) which was published in 1644 that he is regarded as the forerunner of modern Geography. This book was the first attempt to go beyond mathematical geography and mere enumeration of places. *Varenius* was among the first scholars to pinpoint the dualism in geography (i.e. the differences in the character of physical and Human Geography). *Varenius'* book *Geographia Generalis* was the first after the early philosophers to offer detailed theoretical concepts of Geography. The book was not only a summary of what had been accomplished in Geography prior to the time, but it was also a book to look into the future for it contained a theoretical substantiation for the science of geography. It was an attempt to define the objectives of the study of geography and to outline the basic methods for studying the subject. For instance, *varenius* used the comparative method, a method that is widely employed in geographic research.

Geographia Generalis was divided into three parts namely:

- a) The absolute or terrestrial section - this part describes the shape and size of the earth and the physical geography of continents, seas and atmosphere.
- b) The relative or cosmic section - this part deals with all relations and events that occur on the earth through their relations with celestial causes e.g. the influence of the sun on world climates.

- c) The comparative section - which deals with the location of different places in relation to each other and the principles of navigation.

It was *Varenius* who saw the necessity of developing specific branches of geography such as geomorphology, climatology, hydrology etc. *Varenius* argued that Geography is necessary and useful because professionals in other disciplines cannot progress without geographical knowledge.

Perhaps the greatest contributions of *Varenius* to the development of Geography are:

- a) Bringing together contemporary knowledge of astronomy and cartography and subjecting the different theories of his day to sound critical analysis.
- b) Dividing Geography into 'general' and 'special' which later became systematic and regional geography.

Immanuel Kant

Kant is well known as a philosopher but less well known as one of the pioneers of modern geography. Nevertheless, his ideas have had a great influence in the development of geography as a scientific discipline.

Kant is believed to have been the First University teacher to give classes in Geography. He gave lectures in physical geography at the Königsberg Universität. These lectures were significant for the later development of geography.

Kant's influence in Geography unlike that of Varenius was exerted through his students than his writings.

Kant found that geography was a subject inadequately developed and organized during his time. He, therefore, embarked on assembling and organizing materials from a wide variety of sources. *Kant* also considered a number of specific geographical problems e.g. the deflection of wind direction due to earth's rotation.

Kant argued that our knowledge of nature comes from physical geography, consequently geography forms the basis of our knowledge of the world. To *Kant* it was necessary to be acquainted with geography as a branch of knowledge that can be improved and expanded by practical experience.

Like *Varenius*, *Kant* was interested in travels. He argued that interaction with other people through travels broadens the scope of our knowledge.

Kant gave Geography a philosophical foundation. He believed that Geography has a significant scientific contribution to make. He pointed out that empirical phenomena could either be classified according to their nature or according to their position in space and time. *Kant* pointed out that the former is a logical classification while the latter is a physical classification. Logical classification lays the foundations for the systematic sciences, but it is the physical classification that gives history and geography a scientific basis. According to *Kant*, history is a chronological science for it studies phenomena that follow one another

in time. On the other hand, geography is a chorological science for it

studies phenomena that lie side by side in space. To *Kant* history and geography are essential sciences standing alongside the systematic sciences. Like *Varenius*, *Kant* believed that without the knowledge of Geography and History man cannot achieve full understanding of the world.

Kant gave geography a central place among the sciences, and geographers have used his views in justifying the existence of the subject and its special position among the sciences.

Kant perceived geography as a dynamic science. He argued that objects change to provide a different geography at different times. This concept of change in geography developed into the idea of paradigms in geography.

Activity

Compare and contrast the contributions of Bernhard Varenius and Immanuel Kant to the development of geography

Summary

In summary, an attempt has been made to discuss the contributions of Varenius and Kant. Both scholars are regarded as pioneers of modern geography. Their ideas have been of great influence to the development of the subject to its present status.

Definition of key words.

- a) Bernhard Varenius: A medical doctor who later became a geographer. He was very instrumental in the institutionalization of academic geography. He divided geography into general and special.
- b) Immanuel Kant: a philosopher who turned into a geographer. He wrote the book *physische geographie* (physical geography) and divided other geographies into five types:-
- Mathematical geography, in which the shape, size and motion of the earth, and its relationships to the solar system in which it is situated, are dealt with;
 - Moral geography, in which the diverse customs and characteristics of people of different regions are told about;
 - Political geography, whereby the political organization of a state is seen as totally dependent on its physical geography;
 - Commercial geography, concerned with trade linking together areas of excess with areas of deprivation, and
 - Theological geography concerned with the way in which theological principles are transformed due to differences in the land.
- c) *Geographia Generalis* (General Geography); A book written by Bernhard Varenius and divided into three parts:-
- The absolute part, concerned with the body of the earth and its constituents, such as land masses and rivers, together with their properties such as shape, movement and size;
 - The respective or relative part, pertaining to the effects of celestial phenomena on the earth; and the comparative part, which seeks to explain those properties emerging

from a comparison of different places of the earth. Likewise, varenius saw special geography as being concerned with three kinds of things in any region, namely terrestrial, celestial and human.

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LESSONS 16, 17, & 18

GEOGRAPHY DURING THE CLASSICAL PERIOD (17TH - 19TH CENTURY).

Introduction.

In the previous lessons we have looked at Geography during the Renaissance period, noting the contributions of two prominent geographers namely Varenus and Kant to the development of geographical thought. We have noted that these two scholars were among the pioneers of the discipline geography. Their ideas have contributed significantly to modern geography and its present status among the sciences. Both Varenus and Kant devoted great attention to organizing and developing geography which during their times was a discipline inadequately organized and developed. We have further noted that Varenus gave geography its theoretical base while Kant gave it its philosophical foundation.

In these lessons, we are going to look at Geography during the classical period (17th - 19th Centuries). We shall first look at the contributions of two prominent German geographers (namely Alexander Von Humboldt and Karl Ritter) to the development of geographic thought. Then we shall elucidate the Darwinian ideas and examine the impact of Darwin on the development of geography. The French school of Regional Geography under Paul Vidal de la blache will also be examined. Two important concepts in Geographic thought literature namely environmental determinism and possibilism will be introduced and their proponents briefly discussed. A section will follow this on geographic exemplars and their associated attributes. Finally we shall trace the evolution of geography as a paradigmatic discipline.

Objectives

By the end of these lessons, you should be able to:

- a) Discuss the contributions of Alexander von Humboldt and Karl Ritter to the development of geographic thought scholarship;
- b) Examine the impact of Darwin's ideas and concepts to the development of Geographic thought;
- c) Differentiate between Environmental Determinism and possibilism and briefly discuss their proponents;
- d) Describe the attributes of the major geographic exemplars and tentatively assign paradigmatic status to them; and
- e) Trace the evolution of geography as a paradigmatic discipline.

Content

The works of Alexander von Humboldt and Karl Ritter.

Humboldt and Ritter the great masters of German Geography loom across the pages of the history of science. The geography that these two scholars represented was designated as 'classical' because it dominated the foundation period of geography.

Humboldt and Ritter are popularly referred to as the fathers of modern geography on the grounds that their work was seminal and influenced later generations in France and Germany as well as the whole history of geographic thought.

Contributions of von Humboldt to the development of Geography.

Humboldt's significant contribution to Geography is found in his great work, the *Cosmos*. The findings of his Latin American voyages were published in 30 volumes in Paris. In the *Cosmos* Humboldt gave an interpretative synthesis of the Universe. In the first five volumes Humboldt attempted a description of man as part of nature. He partly exhorted the idea of environmental determinism by asserting that man was dependent on soils and natural processes. But on the other hand he exemplified a possibilist idea that man was able to manipulate natural forces through his intellectual capacity.

Volumes 25 and 26 of the *Cosmos* formed one of the first Regional Economic Geographies dealing with the resources and products of a country in relation to population and political

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controls. Humboldt advanced the theory that countries can only

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prosper by making effective use of natural resources. The *Cosmos* sought to provide solutions to questions about interconnections among phenomena grouped together in rich diversity.

Humboldt was the first geographer to use Isothermal lines to connect areas with equal temperature. Noting the pattern of Isothermals from the lines of latitude he also developed the climatological concept of continentality.

In his South American voyages, he for the first time demonstrated the connection between the great South American waterways of Orinoco and the Rio Negro, a tributary of the Amazon, the only natural connection between two great river systems in the world.

On a visit from Peru to Ecuador, Humboldt became the first scientist to carefully measure the temperature and velocity of the cold water current that was washed ashore. This current was for a long time called Humboldt Current but is today known as the Peruvian current.

In an attempt to test some of the theories on the correlation between elevation and temperature, Humboldt took a number of altitudinal measurements and for a period of 29 years held the record for the highest altitude climbed by man by ascending almost to the summit of Chimborazo in Ecuador up to a height of 19,286 feet.

Humboldt also made remarkable contributions to cartography through making maps. In 1832, he made an Atlas that included hachured maps of central America and the United States popularly known as "The Atlas of the historical geography of the United States".

Studies along the Andes provided Humboldt and his companion Bompland with data on plant distribution which he later used for areal analysis. Thus many regard him as the founder of bio-geography.

Activity

Briefly enumerate the major contributions of Alexander von Humboldt to geographic thought.

Contributions of Karl Ritter to the Development of Geography.

Ritter produced one great major work that presented his scholarly achievement. This was the *Erdkunde*, whose essence is the science of the earth in relation to nature and the history of mankind. In the *Erdkunde*, Ritter postulated the role of geography in the study and instruction of other physical and historical sciences. Ritter was greatly concerned about the connection between history and geography. The *Erdkunde* consisted of 19 volumes that covered Africa and Asia.

Ritter made six maps as a supplement to his *Erdkunde* and this can be cited as a significant contribution to Cartography. To his contemporaries Ritter produced many stimulating

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ideas, for instance he coined interesting theories on nations. He argued that

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each nation had its own individuality and was formed by an aggregate of individuals and thus must have its own territory and localized position. This idea is common in modern political geography.

Ritter's studies, which took an anthropocentric standpoint seeking to relate man and nature and to establish the connection between man and his history, has made him alongside his student Friedrich Ratzel to be regarded as the founders of human geography.

Common features in Ritter's and Humboldt's Ideas and concepts

Humboldt's and Ritter's ideas form the foundation of modern scientific geography. They liberated geography from the gazetteer stage that was characterized by mere descriptions of places studied where facts were presented in a disjointed incomprehensive manner. Ritter's and Humboldt's new geography was characterized by emphasis on application of the empirical method in geographic inquiry. For instance, Ritter emphasized that students should progress from observation to observation in search for general laws. Similarly Humboldt emphasized again and again the need for careful observation of nature and the need for careful and precise measurement with the aim of arriving at laws and theories. In fact Humboldt combined concrete expeditionary inquiry into nature with broad generalizations and made profound inferences on the basis of many years of travel and study of the earth's landscape. Both scholars believed that science must be founded on objective descriptions of observed facts rather than on logical propositions like those proposed by Kant. Their contribution to geographic methodology can be underscored in

that they founded the quantitative scientific approach that is one of the hallmarks of modern geography. In a nutshell Humboldt and Ritter saw geography as a Nomothetic rather than an Idiographic science.

In their reference to the new scientific geography, Ritter used the term *Erdkunde* which means Earth science while Humboldt used the term *Erdeschreibung* which means earth's description. Subsequently, they can be exalted for having contributed to the definition and scope of geography and thus distinguishing it from other disciplines. Ritter's *Erdkunde* and Humboldt's *Cosmos* are considered to be two of the greatest scientific works to date.

Ritter and Humboldt stressed the hypothetic - deductive approach that uplifted geography high among the sciences. Not only did they initiate a new methodology in geographic inquiry, but they also introduced new concepts and ideas to geography that are evident in geographic studies today such as Ritter's spatial relations and Humboldt's Isotherms.

Activity

Compare and contrast the contributions of Humboldt and Ritter to the development of geographic thought.

Impact of Darwin's Ideas on Geographic thought.

The year 1859, which saw the death of Humboldt and Ritter was also the year of publication of Charles Darwin's book, The origin of species. Ritter's ideas were gradually replaced by a materialistic scientific philosophy that emphasized natural laws and causality. Many scholars came to believe that Ritter's work was valueless and unscientific because it could not be accommodated in Darwin's concept of struggle and survival. The new scientific method which came to dominate research was the opposite of Ritter's - the inductive approach which sought to witness to God's plan and existence. The method of natural science, the hypothetic -deductive method seeks prime causes by deduction. It sets up hypotheses to see whether they can explain the reality that is observed. The testing of hypotheses leads to the formulation of natural laws. It was agreed among scientists that religion could not provide explanations of natural phenomena, even some theologians asserted that the Bible was not authoritative on scientific questions. Discussions as to whether or not there was a God behind the laws of nature or whether natural laws were a means of development came to be regarded as unscientific.

The increasing belief in the inherent value of science led to a considerable growth in research into the natural sciences. A group of natural scientists with an interest in physical Geography won a place for Geography in academic establishments during the latter half of the 19th. century. In their research they were able to build on the earlier work of von Humboldt and Mary Somerville whose physical geography had been published in 1848. Late 19th. century geographers concerned themselves more and more with pure physical geography where man played no part. There was a tendency to consider physical geography

as a branch of geology. Late 19th. century geographers also established

geomorphology, the study of landforms, which later became the most substantial element in physical geography.

In Germany Ferdinand von Richthofen and Oscar Peschel led this new direction whereby physical geography was over emphasized. Like Ritter, Peschel was interested in the significance of landforms and the historical development of mankind. Peschel believed that geography should be a systematic, empirical brand of knowledge with its main emphasis on the study of landforms.

In Britain Geography as a distinct field of study was advanced by people like Captain James Machonochie at University College, London and Sir Halford J. Mackinder at Oxford University.

In the United States, Individual University teachers of Geography included William Morris Davis of the Davisian cycle of Erosion fame. Davis, like most of his professional colleagues in other countries was a physical geographer.

In Britain Thomas Henry Huxley's book entitled physiography published in 1877 was very influential in changing the teaching of geography at all levels. At this time the term Geomorphology had not yet been invented, only morphology was in general use. Physiography may be defined as the description of nature, encompassing the systematic sciences of Zoology, Botany and Geology. Huxley's work gives a clear demonstration of

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the post Darwinian causal reasoning. The idea that geography can only be learned

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through local studies, field courses and excursions is to a large degree derived from the educational principles of Huxley.

During the last third of 19th, physical geography (renamed physiography after 1877) became a very popular school subject. Physiography was regarded as an integral part of geography. However, physiography slowly vanished from the syllabuses as more specialized earth sciences became established. physiography was replaced by geomorphology. The American William Morris Davis (1850-1934) became the leading personality in the development of geomorphology, although he preferred to use the term physiography. Davis' organising principle was not simple causality but rather the idea of regular change of form through time, as systematized in his scheme of the cycle of erosion. In Germany, Albrecht Penck made important contributions to Geomorphology through his ideas on peneplanation and pediplanation.

The development and success of physical geography, variously termed as physiography, morphology or geomorphology, led the German George Gerland to suggest, in 1887, that the study of cultural phenomena should be separated from geography.

Developments in physical geography provided the main innovations in geography during the latter part of the 19th. Darwin's work and his influence on the development of natural sciences also influenced geographical research. The following themes from Darwin's work can be traced in geographical studies:

- a) Change through time or evolution, a general concept of gradual or even transition from lower or simpler to higher or more complicated forms. Darwin used the terms evolution and development essentially in the same sense.
- b) Association and organization - man as part of a living ecological organism;
- c) Struggle and natural selection.
- d) Randomness, or chance character of variations in nature.

The concept of change over time is, for example, found in Davis' cyclic system for the development of a landscape through the stages of youth, maturity and old (senile) age, which Davis described as evolution. The French school of regional geography, led by Paul Vidal de la blache, also took up the study of evolution within the cultural landscape. The ideas of association and organization that Darwin had inherited from earlier philosophers and scientists have been widely used in geographical research.

Activity

1. Enumerate the four themes that are taken to be significant contributions to geographic thought from biology, and particularly from Darwin.
2. Briefly explain the contributions of William Morris Davis and Albretch Penck to the development of Geomorphology

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Environmental determinism and possibilism.

Activity

By observing the immediate environment briefly comment on the relationship between the physical and the human environment

The inter-relationship between the physical environment and the human environment has pre-occupied geographical thinking since time immemorial. The relationship between man and nature has been of prime interest to geographers. Geographers have been investigating how the physical environment affects the functioning and development of societies. Debate has been strong as to whether man's activities are influenced by the physical environment or whether man manipulates the physical environment for his development. This debate among geographers has led to the emergence of two schools of thought concerning the relationship between the physical and the human environment. The first school of thought advocates the highest level of physical control over the human environment. This school of thought is referred to as environmental Determinism. The second school of thought advocates the primacy of human control over the physical environment. This school of thought is referred to as possibilism.

Proponents of Environmental Determinism and possibilism and some Geographic Exemplars.

After Darwin and his ideas on survival for the fittest, Geographical research was primarily concerned with recognizing the laws of nature. In Geography, this led to a rather deterministic view of struggle and survival. Nature was studied especially by the physical geographers, with open eyes, seeking as objectively as possible to identify the forces or processes that govern the formation of valleys, uplands and coastlines. Man's achievements were explained as a consequence of natural conditions.

The British philosopher Herbert Spencer who elaborated on the concept of social Darwinism believed that human societies closely resembled animal organisms and that human societies must struggle in order to survive in particular environments much as plants and animals do.

Activity

To what extent is the concept of social Darwinism applicable today?

Ratzel and his Anthropogeographie

The latter half of the 19th century was dominated by the works of Friedrich Ratzel. Most of Ratzel's ideas on human geography were published in 1882 under the title *anthropogeographie*. In his first volume of *anthropogeographie*, Ratzel expounded on his deterministic ideas.

With a doctorate in Zoology, geology and comparative anatomy, he viewed geography as "the connection between the natural sciences and the study of man". He advocated the use of scientific methods in the study of human geography. One of Ratzel's greatest contributions was to have brought this part of the geography of man (cultural geography) into a scientific system by organisation of the phenomena and the establishment of concepts and significant connctions of the results obtained. In *Anthropogeographie* Ratzel used the deductive approach to present the first systematic study of the geography of man. *Anthropogeographie* vol. 1 was organized largely in terms of the natural conditions of the

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earth, which Ratzel studied in relation to human culture. The

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environment - dominated - man - response approach (environmental determinism) was reversed in his second volume of *anthropogeographie* that emphasized the possibilist way of thinking. This possibilist way of thinking, with useful modifications by Hettner and Schlüter influenced the development of Richthofen's chorological approach, an approach that became associated with 20th century German geography. Elsewhere, especially in the U.S, Ratzel's earlier viewpoint in the 1st edition of *anthropogeographie* was extended and elaborated by scholars such as Huntington, Semple and Davis into what became known as environmental determinism; a view which dominated American geography for decades. As Grossman (1977) observed:

"The scientific milieu in the latter half of the 19th and early 20th centuries was dominated in part by Darwinian ideas, deductive approaches and an acceptance of the concept of Newtonian cause and effect relationships. Fitting well into this intellectual environment, the theme of environmental determinism, developed mostly by geographers, was the prevailing view in American Geography at the turn of the 20th century... The concern was with documenting the control or the influence of the environment upon human society."

In America, environmental determinism was very widespread. William Morris Davis, for example, stated that geography was concerned with the analyses of the relationships between inorganic control and organic response.

Semple argued that man like other organisms was a product of the earth. Huntington argued that weather and climate influenced the trends in human history. Central to all these works on the effects of the inorganic controls on organic response was the notion of Newtonian cause and effect relationships.

The influence of Ratzel was not just restricted to the initiation of the environmental deterministic approach, he made contributions in the field of political geography. Concepts such as *lebensraum* (German word for living space), and the organismic structure of the state featured prominently in his *politische Geographie* (political geography) published in 1897. Ratzel also made contribution to historical geography. Sauer refers to Ratzel's *culturgeographie* as 'historical geography of grand design'.

In a nutshell, Ratzel's *anthropogeographie* was a seminal work and the amount of intellectual debate it created on both sides of the Atlantic makes it an exemplar. Ratzel's views about geography dominated for decades -a tribute to his ability as a teacher and scholar. Friedrich Ratzel and his teacher Karl Ritter are widely recognized as the founder of human geography.

Activity

Itemize the contributions of Friedrich Ratzel to the development of geographic thought scholarship.

Paul Vidal de la Blache and the French school of Regional Geography

Equally central to the development and evolution of geography during the early part of the 20th century was the work of the French geographer Paul Vidal de la Blache. Unlike the inorganic control- organic response basis of the environmental determinists, Vidalian tradition minimized the influences of the environment. Vidal together with Lucien Febvre (a French historian) are widely regarded as the pioneers of the possibilist way of thinking. Originally, a historian, Vidal read the works of Humboldt and Ritter, and later tested his ideas on human geography against those of Ratzel. Central to Vidal's work were the lifestyles (*genres de vie*) that developed in different geographic environments. Vidal's concept of natural milieu, which evolved from his studies of rural communities, stressed the relationship between the inorganically integrated and biotic infrastructure of human life on earth.

Vidal believed that social geography should concentrate on understanding how both the biotic and physical conditions become manifest in the social life of the various societies. In explaining the differences between groups in the same or similar environments, Vidal relied not on the dictates of the physical environments, but on the attitudes, values, and habits of the communities. He argued that nature sets limits and offers possibilities for human actions. Vidal strongly argued for possibilistic viewpoints and contrasted them sharply with deterministic views presented by Ratzel in his first edition of *anthropogeographie*. In the study of these human possibilities, Vidal advocated a holistic

approach: the geographer should study the whole and show that every region is a composite of mutually interdependent parts.

Vidal (1848-1918) is regarded as the founder of modern French geography. Vidal emphasized the study of regions that are unique. He argued for regional geography and against systematic geography as the core of geographical studies.

Possibilism, developed from the works of Vidal and Febvre, continued to grow and spread on both sides of the Atlantic. In France, the major proponents and refiners included Jean Brunhes whose unfoundering support to the possibilist viewpoint and empiricism is well documented. It was Bruhnes who enunciated the first explicit formulation of human geography as a systematic approach to the study of geography.

Outside France, the possibilist ideas, in contradistinction to environmental determinism, were accepted by a large group of geographers and anthropologists. In America, the possibilist viewpoints were advanced by Barrow and Carl Sauer.

Sauer and his morphology of the Landscape

Sauer's ideas well exemplified in his papers and in the book 'morphology of the landscape' were regarded as significant extra ordinary contributions to geography. Sauer rejected the paradigm of areal differentiation as put forward by Vidal, asserting that the geographer's role is to investigate and to understand the nature of the transition from the natural to the

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cultural landscape, and the successive stages through which the cultural

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landscape has passed during its transformation. From such an exercise, the geographer would identify the major changes that have occurred in an area as a result of occupancy by a succession of human groups.

Because of the difficulties in the analysis of the transition from the natural to the cultural landscapes, Sauer had to modify his original goal of understanding the transition from natural to cultural landscapes. He developed a historical framework for the study of landscape development with a focus on the patterns of human occupancy rather than on the socio-cultural agencies that generates the patterns. In his studies Sauer's emphases were on occupation and land use. He argued that the dimension of time, is and has been part of geographic understanding. According to Sauer Human geography considers man as a geographic agent, using and changing his environment in non-recurrent time according to his skills and wants.

Historically, Sauer was influenced by many European geographers, especially Otto Schlüter and Siegfried Passarge. Schlüter believed that the distinctiveness of a region could best be studied by a reconstruction of the original landscape, the *Urlandschaft*, followed by a historical account of the stage through which the landscape was transformed by man, the *Kulturlandschaft*. Slightly different from Schlüter, was the view of Passarge that the landscape was not unique, it was a type. Passarge saw a landscape type as what we would call a spatial system, an assemblage of interrelated elements.

Note

Sauer's contributions to the evolution of possibilism and American geography have been quite immense

Activity

Question

Briefly explain the contributions of Sauer and Schluter to the development of the Possibilist way of geographical thinking.

Hartshorne and his Nature of Geography

Chorology as an approach to the study of geography was first developed by Richthofen and elaborated and refined by Alfred Hettner in a series of papers published in *Geographische Zeitschrift*. In America, the chorological viewpoint was proposed and defended by Richard Hartshorne. Hartshorne subscribed to the Areal differentiation paradigm and pointed out that the goal of the chorological point of view is to know the character of regions and places through comprehension of the existence together and interrelations among the different realms of reality and their varied manifestations, and to comprehend the earth's surface as a whole in its actual arrangement of continents, larger and smaller regions and places.

Hartshorne, like Hettner, argued that geography could use both the idiographic and nomothetic (nomological) approaches, although scholars, like Bunge and Harvey argue that the use of place by Hartshorne implied an idiographic approach. Hartshorne advocated the scientific approach to geography and this was in part, responsible for the

difficulty in classifying his position on the idiographic/nomothetic debate. Hettner's work published in *Geographische Zeitschrift* constituted the initial exemplary views on chorology. Richard Hartshornes book, *The Nature of Geography*, (1961) codified the chorological viewpoint into one of areal differentiation. Hartshorne used the term areal differentiation to characterize the way in which geographers dealt with the wide variety of phenomena, (physical, economic and social) which exist together in an area and distinguish them from other areas.

Although Hartshorne's book, *The Nature of Geography*, has been criticized over the years, it is still, a central textbook in courses on geographic thought. The book has provided, for decades, research foci for geographer all over the world. *The nature of Geography* is a book that provides a synthesis of the works of European and American geographers. It also provides a geographic research viewpoint that advocates regional studies emphasizing areal differentiation. Most scholars argue that if one seeks to understand recent developments in geography one must begin with an examination of *The Nature of Geography*.

Activity

Go through Hartshorne's book 'The Nature of Geography' and write down the major Geographical ideas presented in that book.

Schaefer's Exceptionalism in Geography

The 1950s and 1960s were periods of discontentment with the subject matter of geography. The widespread acceptance of quantitative methods made changes in geography as a discipline inevitable. Geographers started to have a spatial bias in the selection of geographic problems. Schaefer advocated for a geography that is more nomothetic and based on spatial theories. This marked the inception of the spatial viewpoint in geography. Schaefer advocated the use of spatial laws as a basis for geographic explanation.

Ullman restated the spatial viewpoint advocated by Schaefer. Ullman suggested that geography should focus on spatial interaction. According to Ullman geographers should concern themselves with space and spatial interaction.

The untimely death of Schaefer deprived Geography as a discipline of further insights into his thoughts about spatial laws and geographic explanation. Thus, unlike the other exemplars that benefited from further elaboration's by the authors, the spatial viewpoint had to develop through the articulations of future scholars. Bunge's *Theoretical Geography* (1962), Haggett's *Locational Analysis in Human Geography* (1966) and David Harvey's *Explanation in Geography* (1969) are basically elaborations of Schaefer's original criticisms and suggested modifications to Hartshornian orthodoxy. Largely because of the concerted intellectual contribution to the spatial theme, the major tenets

have gradually evolved; from Ullman's spatial interaction theme to the theme of spatial organization which was institutionalized by Abler Adams and Peter Gould (1971). Subsumed under the umbrella of spatial organization are numerous studies on spatial interaction, spatial systems, and social planning and regional taxonomies. These studies emphasize both spatial patterns and spatial process.

Note

In the discussion of what we regard as exemplars in geography we have generally focused on those works that have had profound pedagogical and methodological import in the discipline of geography. For a discipline that has evolved for over a century, it is of course, difficult to discuss the works of all those scholars that have had an impact upon the field. In the above we were only interested in those works we regard as initiating significant changes in the focus of the field over an appreciable period of time. The major geographic exemplars and their associated attributes are summarized in the table below.

Geographic exemplars and their associated attributes.

Author	Exemplar	Image of subject matter	Theories & laws	methods	Area of discontent
Friedrich	Anthropogeogr	Inorganic	Darwinism	Deductive	Too much

Ratzel	aphie	control-organic response	(stage theories of physical and social systems) Environmental determinism	newtonian cause and effect systematic approach	environmental control, Absence of Human initiatives and choice: eliminated novelty & creativity.
Paul Vidal de la Blache	Tableau de la Geographie de la France	Changes in attitudes, values and habits create possibilities for human communities	Genres de vie: the product & reflections of the inter-relationship between man and his environment	Fieldwork/case studies Emphasis on causal successions or sequences.	Too much emphasis on a region
Carl Sauer	The four 1919-1922 papers morphology of landscape agricultural origins & dispersal's The Early Spanish man	Time-the fourth dimension in geography landscape view of geography	Cultural landscapes evolve from the physical landscape man was an agent of environmental modification	Fieldwork and historical reconstruction of the human landscape inductive development of landscape patterns.	Pre-occupation with pattern rather than process. The inability to analyze societal values, beliefs and social organization
Richard Hartshorne	Nature of geography perspectives on the nature of geography	Chorology perceived as idiographic	Functional relationship order-classifications	Fieldwork mapping	No laws No generalization too restrictive a view.
Schaefer	Exceptionalism in Geography	Spatial interaction spatial organisation nomothetic appeal	Location flows distribution settlement	Mathematics statistical methods scientific method	The method is too restrictive and value less.

Geography as a paradigmatic Discipline.

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Definition of a paradigm

Question

What do you understand by the term paradigm?

Since 1972 the use of the word paradigm has become fashionable in geography. The term is a pivotal concept for courses in geographic thought on both sides of the Atlantic. Thomas Kuhn in his classic book 'The structure of scientific revolutions' defined paradigm as "the entire constellation of beliefs, values, techniques and so on shared by members of a given community." The term paradigm has been widely used in social science research but without any consistent meaning. It refers to a body of ideas (or a broad model) which may be used to guide the development of theory and explanatory research. Scientists whose research is committed to the same problems, rules and standards often share such a body of ideas. Kuhn (1962) defines paradigms as "universally recognized scientific achievements that for a time provide model problems and solutions to a community of practitioners.

Hagget (1979) sees a paradigm as a kind of a super model or a superconcept. Kuhn used the paradigm concept in at least 21 different ways which were later collapsed into three paradigm types by Masterman (1970) viz: the metaphysical paradigms or metaparadigms; the sociological paradigms and the artefact or construct paradigms. The metaparadigms present a total global view of science. It is a gestalt view or a map. Such a map allows the scientist to explore the vast and complex world. Such a paradigm performs three basic functions:

- a) it defines what entities are (and are not) the concern of a particular scientific community;
- b) It tells the scientist where to look (and where not to look) in order to find the entities of concerns to him.
- c) It tells the scientist what he can expect to discover when he finds and examines the entities of concern to his field. A metaphysical paradigm is the broadest area of consensus in a discipline, and it defines the sub-areas of research. In contrast, the sociological paradigm is grounded on concrete scientific achievement; a universally recognized scientific achievement. The third and narrowest, is the construct paradigm. Here specific entities such as a textbook, and instrument or a classic work (an exemplar) are viewed as paradigms. Conceptually, the construct paradigm is largely subsumed in the sociological paradigm, and this in turn is essentially subsumed in the metaparadigms. Because of such a nesting, the construct paradigm is central to the development of paradigmatic sciences. Construct paradigms are problem - solving paradigms.

In geographic literature, many scholars have assumed that geography is a paradigmatic science. Geography as a discipline has undergone various metamorphoses, revolutions that signal the demise of one paradigm and the creation of another. A central focus to the formation of a paradigm is the existence of an exemplar (classic work) or exemplars. The discussion of exemplars above suggests that paradigms may have developed around them. A viewpoint may become a paradigm, if in addition to an exemplar, it has a large following and if it has clearly defined theoretical and methodological bases.

The exemplars as outlined in the table above had significant following and we can tentatively assign paradigmatic status to them:

- a) Ratzel with the paradigm of environmental determinism;
- b) Vidal with that of possibilism
- c) Sauer with the landscape paradigm
- d) Hartshorne with the chorological paradigm which he later codified into the Areal differentiation paradigm and
- e) Schaefer with the spatial organisation paradigm.

In a historical framework, Ratzelian determinism was partly contemporaneous with vidalian possibilism, Sauer's historico-anthropological paradigm was basically contemporaneous with Hartshornian chorology. The modifications of both chorology and the landscape view gave rise to the spatial organization paradigm.

Note

The above paradigmatic framework is only valid if we abandon the Popper-Kuhn view that a new paradigm is created as a result of a revolution. Indeed, Ratzelian determinism and Vidalian possibilism were very different and extensive debates did occur between their numerous disciplines, but there was no real revolution prior to the emergence of possibilism. When two paradigms co-exist simultaneously, the discipline is a dual paradigmatic discipline. Similarly, the historico-anthropological and the areal differentiation viewpoints may have been divergent, but one did not emerge as a result of a revolution; a dual-paradigmatic discipline was still operational.

Note

It should be noted that the dual-paradigmatic period we have associated with the Sauer-Hartshorne viewpoints does not imply that environmental determinism and possibilism, as per Ratzel and Vidal were defunct. It implies, however, that their dominance as research and pedagogical vehicles had diminished. This tendency for older paradigms to persist beyond their dominant periods was particularly true of the 1960s and 1970s when the spatial organization paradigm was in vogue. Concomitantly, The anthropological-historical approach of Sauer and the areal differentiation approach of Hartshorne were still operational.

The spatial organization paradigm was dominant during the 1960s and 1970s and continues to be important. This spatial paradigm was capable of such general acceptance because it was initially packaged in the framework of a specific philosophy, logical positivism. Indeed, the book, 'Explanation in geography' was a timely vehicle for linking the spatial paradigm to laws and theories through rigid mathematization. The resource paper series of the Association of American Geographers (AAG) and the concepts and techniques in Modern Geography (CATMOG) series of the Institute of British Geographers were suitable vehicles for conceptual, theoretical and methodological dissemination of the spatial viewpoint. Advancement in the use of computers also solidified the spatial organization paradigm. Books such as 'Spatial Organization' (by Abler Adams et. al, 1971) and 'Models in Geography' (by chorley and Hagget, 1967) were timely for the consolidation of the spatial viewpoint and the emergence of a single paradigm.

Note

If the revolutionary criterion for a new paradigm development is relaxed, then geography has evolved from being a dual-paradigmatic discipline to a single paradigmatic discipline. However, if the Kuhn-Popper criterion is maintained, then only the spatial organization paradigm may have evolved as a result of a revolution. For a theory or paradigm to be accepted, it must possess certain basic characteristics:

- It has to attack the main theme of the prevailing orthodoxy;
- The theory must be new, yet it must incorporate some components from the existing orthodoxy
- It must be relatively difficult to understand
- It must offer a new methodology and
- Finally it must offer important empirical relationships for measurement and analysis.

Of all the paradigms discussed above only the spatial one seems to meet these criteria. By the very nature of the exemplars discussed, geographic paradigms are examples of artefact (construct) paradigms. Construct paradigms, as noted earlier, may be based on artefacts such as a book, they may be sub-theory and their development may not necessarily fit the Kuhn-Popper revolutionary process. Social sciences such as Geography may change by extension and reorientation rather than by revolution. Consequently, geography has evolved as being initially a dual paradigmatic discipline towards a single paradigmatic science in the 1960s and 1970s.

Activity

1. Identify the major geographic exemplars and describe their attributes. Also assign paradigmatic status to them.
2. State the major paradigms in geography and citing specific examples argue the case for the one that flourishes today in modern geography.

Summary

The first part of these lessons introduced you to the works of two prominent German geographers namely von Humboldt and Karl Ritter. We noted that these two are widely regarded as the founders of modern geography. Von Humboldt published his work in the *Cosmos* while Ritter's scholarly achievement is published in the *Erdkunde*.

The second part of these lessons examined the impact of Darwinian ideas to the development of geographic thought. We noted the four themes that are taken to be significant contributions to geographic thought from biology and particularly from Darwin. These themes are: change through time or evolution, Association and organization, struggle and natural selection and randomness or chance character of variations in nature.

The third part introduced two important concepts in geographic thought literature i.e. environmental determinism and possibilism and looked at their proponents.

In the third part we looked at geographic exemplars and their associated attributes. The exemplars we looked at included:

- Ratzel's arthropogeographie;
- Vidal's Tableau de la Geographie de la France;

- Sauer's morphology of Landscape;
- Hartschorne's nature of Geography and
- Schaefer's Exceptionalism in Geography.

After identifying the above exemplars we defined the term paradigm and tentatively assigned paradigmatic status to these exemplars.

The following paradigms were identified:

- Environmental determinism;
- Possibilism;
- Landscape paradigm;
- Chorological paradigm;
- Areal differentiation and
- Spatial organization paradigm.

Definition of key words used in the Lesson.

- *Cosmos* (meaning the universe): this is the great work written by Alexander von Humboldt.
- *Erdkunde* (means earth science): this is the book in which Karl Ritter has presented his scholarly achievement.
- Gazetteer geography: A Geography characterized by mere descriptions.
- Nomothetic: a concern with the general and universal: the opposite of idiographic.
- Idiographic: A concern with the unique or particular: the opposite of nomothetic.
- Deductive: the derivation of a particular truth from a general statement. It is a form of inference through which some premises and propositions are taken as basic, and all

other propositions are seen as logically followings from them. The opposite of inductive.

- Inductive: Argument through which general statements are derived from particular ones. More formally, it is used to refer to arguments, in which the truth of premises provides a reasonable basis for belief in a conclusion, although they do not actually entail the truth of such a conclusion. The opposite of deductive.
- Materialism: Philosophy which gives primacy to the material world, and which in contrast to idealism, relegates mental constructs to a secondary or dependent position. Marx's historical, or dialectical materialism sought to explain the dynamics of social change by focusing on the material foundations of social life, termed the economic base or mode of production. Changes in mode of production are seen to arise through tensions or conflicts in the relations of production of an earlier mode.
- Environmental determinism: A school of thought or a paradigm that advocates the highest level of physical control over the human environment. The opposite of possibilism.
- Possibilism: A school of thought or a paradigm that advocates the primary of human control over the physical environment. The opposite of environmental determinism.
- Paradigm: According to the Popper - Kuhn definition, paradigms are the universally recognized achievements that provide model problems and solutions for members of a scientific community. A paradigm is taken to define both the problems and the methods used to solve them.
- Chorology (or chorography): the investigation of the areal differentiation of the earth. Ptolemy distinguished between geography, the study of the earth as a whole, and

chorography, the description of selected regions of the earth. This difference was paralleled by Varenus' distinction in the 17th century between general and special geography, and more recently between systematic and regional geography.

Further Reading

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LESSONS 19, 20, & 21

NEW TRENDS AND METHODS IN GEOGRAPHIC RESEARCH AND TEACHING.

Introduction.

In the previous lessons, we have looked at Geography during the classical period. In particular, we have noted the works of Alexander von Humboldt and Karl Ritter who are widely regarded as the founders of modern geography. We have also looked at some geographic exemplars and their associated attributes. We have further defined the term paradigm noting the various types of paradigms. Finally we have assigned paradigmatic status to the geographic exemplars.

In these lessons, we are going to look at new trends in geographic research and teaching. We shall first answer the question whether geography is an idiographic or a nomothetic science. We shall then look at the positivist movement and the quantitative revolution trying to see how they have affected geographic research and teaching. We shall finally briefly look at the radical revolution and see how it has affected the subject matter of Geography.

Objectives

By the end of these lessons, you should be able to:

- a) Distinguish between Idiographic and Nomothetic approaches in geography;
- b) Explain the effects of the positivist movement and the quantitative revolution on geographic research and teaching; and

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c) Briefly discuss the effect of the radical revolution on the subject matter of geography.

Content

Question

Is geography an idiographic or a nomothetic science?

The definitions of geography by a number of scholars (Hartshorne, Yeates etc) clearly portray that the question as to whether geography is a nomothetic or an idiographic science is one of methods rather than content. Nomothetic here refers to a method that is law seeking and is concerned with the establishment and verification of generalizations while idiographic is a method which stresses the individuality and uniqueness of phenomena as opposed to their similarities.

Geography, which traces its name to the Greek scholar Eratosthenes who lived in the 3rd Century B.C., has had an interesting history characterized by shifts in methodological approaches. Therefore to ascertain the place of geography in the Idiographic - Nomothetic debate one needs to unveil its history. Going back to the ancient Egyptian and Greek world where Geography was founded, we realize that scholars then used a purely descriptive (idiographic) method, in their study of places and other geographical phenomena. Greek, Egyptian, Roman and Muslim scholars gave mainly accounts of place names, commonly referred to as gazetteer geography. To mention just a few of the Greek scholars we have anaximanders (610 - 546 B.C.), Anaximenes (552-525 B.C.) Hecateaus and Thales (624-547 B.C.). Others include Herodotus and Strabo (64 B.C.- 22 A.D.) who is said to have distinguished between general and regional geography. The Roman

scholar Claudius Ptolemy emphasized chorography, the science that gives detailed accounts of individual regions with their noteworthy features. Though some of the Greek scholars like Thales did actually base their work on the rudiments of scientific knowledge, theirs cannot be qualified as systematic (nomothetic) because they never applied the hypothetic - deductive method that is law seeking.

The nomothetic - idiographic debate in the dawn of modern geography was triggered by Bernhard Varenus (1622 - 1650) who divided geography into 'general' and 'special' geography. General geography dealt with the world as a unit which could be understood through natural laws while 'special' geography was intended as the description of individual countries and world regions.

The fathers of modern geography, Alexander von Humboldt (1769-1859) and Karl Ritter (1779 - 1859) opposed the idiographic approach. They believed that geographers should collect empirical data and should later shape, order and form associations from this material in order to arrive at deductive explanation. Humboldt and Ritter advised that a time had reached in its history when geography should behave like other sciences and stop being content with mere descriptions.

But the classical phase with its nomothetic bias did not go far before facing an overthrow by regionalism, an approach founded by Paul Vidal de la Blache (a French Geographer). Vidal advocated an idiographic approach in studying the cause-effect relationship. The idiographic view continued to gain wide currency between the first and second world

wars under the umbrella term Landscape or *landschaft* geography. Landscape means "a scientifically defined geographic region" a concept coined by the American geographer Carl Sauer. Following Sauer's paper of 1925 'morphology of the landscape' geographers started studying regions (Landscapes).

The idiographic method continued to act as the dominating paradigm till the 1950's. Perhaps no geography scholar conditioned the idiographic approach as did Hartshorne in his book 'The Nature of Geography'. Hartshorne viewed geography as the study of areal differentiation. The proponents of the idiographic approach and regionalism emphasized that the unique cannot be generalized and it is therefore a fallacy to establish theories when studying spatial phenomena.

However, the face of geographical methodology started changing in the late 19th century and geographic research went nomothetic. The aspects of Darwinism came to the limelight. The hypothetic - deductive methods were applied. Inductive arguments were increasingly replaced by the deductive approach. However, despite this systematic tendency in Geography, Geographers for a long time continued to emphasize the central position of regional geography in the discipline. Chabot as quoted by Holt - Jensen (1981) declared that "regional geography is the center around which everything converges". At this juncture it seems like geography was now sailing in what Jensen (1981) refers to as a dual-paradigm; some attempting a systematic approach and others conserving the idiographic approach; albeit, some geographers found it difficult to break

from the old traditions developed by Hartshorne. Regional geography flourished in France where geography was closely associated with history in schools and Universities.

It was not until the 1950s that geographers openly expressed their dissatisfaction with regional geography with its idiographic bias. Schaefer (1953) for instance, in his book *Exceptionalism in Geography*, criticized Hartshorne's claim of uniqueness of geography in methodology. Schaefer maintained that objects in geography are not more unique than objects in other disciplines. He thus prescribed the kind of laws that geographers ought to seek, and urged them to study systematic geography.

Consequently from the 1950s to 1960s there was a movement towards making geography scientific (nomothetic). This trend was referred to as the positivist movement while others called it the quantitative revolution. The quantitative revolution was characterized by the use of hypothetic - deductive method that seeks to establish laws and theories in determining causal effects. Use of mathematics and statistical methods was emphasized for the explanation and prediction of phenomena. Model building came to the fore. A model was defined as an idealized or simplified representation of reality.

The quantitative revolution lifted geography high up among the sciences. Geographers now sought principles and generalizations, which is the aim of all sciences. With the nomothetic approach geography could easily be associated with other scientific disciplines. By the mid 1960s, quantification had taken root especially in Western Europe and North America.

Surprisingly enough, though most supporters of the quantitative revolution felt that geography had achieved its ideal methodology hardly did a decade elapse before the new geography started facing criticisms and challenges. Quantification was disillusioning since it was not leading to substantial formulation and development of theory. People started wondering whether models could explain human behavior. Stamp (1966) felt that over emphasis on law seeking methods may tempt the geographer to discard important data. Sack (1972) maintained that space, time and matter cannot be separated analytically in a science that is concerned with providing explanation.

In a nutshell, positivist geography was criticized for concealing real life situations.

In the 1970s and 1980s geographers got dissatisfied with the generalizations they had produced through the nomothetic approaches. Harvey (1973) noted the following about the quantitative revolution "Our paradigm is not coping well ... it is ripe for overthrow". As a counter - thesis to the nomothetic trend, several schools have come up in the development of geography, in what Holt-Jensen (1981) calls a new 'critical revolution'. These include behaviourist and humanistic approach and the welfare approach. Most significant has been the radical revolution concerned with social issues. Radical geography emphasizes on an orientation that is able to respond to the major contemporary social issues with particular target on pollution, poverty, hunger, injustice and exploitation of colonial territories by capitalist nations.

Several geographers have suggested that we should go back to the idiographic orientation.

Others have suggested an integrated approach of both the nomothetic and

idiographic approaches. They believe that systematic and regional geography is inseparable.

Activity

Precisely place Geography in the Idiographic- Nomothetic debate

Geographic Thought in the 1970s and 1980s

Positivism and the quantitative revolution

As noted earlier, the spatial organization paradigm was associated with a specific philosophy - positivism. From 1963 up to the beginning of the 1970s, the positivistic approach was virtually unchallenged and David Harvey's book 'Explanation in Geography' was an attempt to establish the new geography on stronger philosophical foundations.

Although the spatial viewpoint was not directly challenged, the initial criticisms against this viewpoint were directed against its philosophical, methodological and theoretical bases. By its very nature, positivism is concerned with aggregate patterns, with the explanation and prediction of spatial patterns. In these ventures man is portrayed as rational and his spatial behaviours as reflections of an organism that follows spatial strategies which maximize some subjective utility function. Consequently, in developing theory human geographers, like economists have relied heavily on a priori or normative models, which are based upon assumptions of perfectly rational people, isotrophic planes and the like.

Note

In tracing the evolution of evolution of geography as a paradigmatic discipline one should note the traditions of man - environment relations, (environmental determinism and possibilism)); regionalism (chorology and areal differentiation), spatial organization, behaviouralism (subject - humanistic and dialectical (marxist) approaches. All these paradigms are present in today's geographic literature, which reflect the continuity of these paradigms. Such a variety is a sign of a vigorous field of scholarship and to reduce them to a single paradigm (or superconcept), would be misleading. Nevertheless, most geographers would agree that the spatial organization paradigm flourishes in geography today. Consequently most courses in geography have made the spatial organisation paradigm the main basis for organising and presenting material. The spatial organisation paradigm in modern geography provides a framework for analysing location of activities and the spatial structure and behaviour of human activities. This paradigm was very much influenced by the quantitative revolution and positivism philosophy.

The Quantitative revolution:

Since the 1960s geography has undergone a radical transformation of spirit and purpose, best described as the quantitative revolution. The consequences of the revolution involve the mathematization of geography with an attendant emphasis on the construction and testing of theoretical models. Physicists and mathematicians began this movement that led to the revolution in geography.

The two main elements of the quantitative revolution are:

- a) Quantification and experimentation with a wide range of statistical techniques and
- b) Emphasis on the formulation of hypotheses and search for theory.

Note

Geographers found the theories they were looking for not in geography but in economics and the biological and behavioural sciences including location theory and the general systems theory.

Quantitative techniques are appropriate methods for the development of theory in geography. The quantitative era will last as long as its methods can be shown to be aiding in the development of theory, and there can be no end to the need for more and better theory. It follows that any branch of geography claiming to be scientific has need for the development of theory, and any branch of geography that has need for theory has need for quantitative techniques.

Positivism.

Positivism, in some connections called empiricism has the central thesis that science can only concern itself with empirical questions (those with a factual content) and not with normative questions (questions about values and intentions). Empirical questions are questions about how things are in reality. In this context 'reality' is defined as the world which can be sensed. This means that science is concerned with objects in the real world. Empirical questions are about what a thing is, but normative questions are about how a thing should be. How are the available food resources distributed among Kenyans? Is an

empirical question. The corresponding normative question would be 'how should the available food resources be distributed among Kenyans?

Positivism emphasizes on the unity of science. Scientific status is guaranteed by a common experience of reality. A common scientific language and method ensures that observations can be repeated. Since science has a unified method, there can only be one comprehensive science. The common method is the hypothetic - deductive method.

The Radical Revolution:

During the late part of the quantitative revolution, geography experienced a second revolution referred to as the radical revolution. This revolution made geography to begin emphasizing contemporary issues related to human problems in the environment. This new approach called the welfare approach gave birth to social geography and medical geography. American geographers pioneered this new approach with the aim of giving geography more social relevance.

Activity

Examine the main elements of the quantitative revolution and positivism showing how they have affected geographic teaching and research.

Summary

In these lectures, we have addressed the question whether geography is an idiographic or a nomothetic science. We have seen that each of these approaches has had a place in the history of geographic thought. However, we have noted that the nomothetic approach

flourishes today in geography. We have also looked at the positivist movement and the quantitative revolution. These two led to the introduction of statistical techniques, formulation of hypotheses and computer analysis in geographic research. We have finally touched on the radical revolution (Welfare approach) noting that this approach has led to geography focussing on contemporary human problems in the environment.

Definition of key Words

- Quantitative revolution: A revolution started by physicists and mathematicians and led to the mathematization of geography. The revolution also emphasizes on the formulation of hypotheses, model building and search for theory in geography.
- Positivism: A 19th century philosophical movement originating in the work of saint - Simon, and developed by Comte. Its central claims are that science is the only valid form of knowledge, and that observable facts, are the only possible objects of knowledge. Built on the assumptions of phenomenism and empiricism it is directly opposed to meta-physics.
- Theory: logically constructed statements designed to provide explanation or understanding, but which vary greatly from philosophy to philosophy.
- Law: a general statement accepted as universally true. In positivism, laws are produced through the empirical testing of hypotheses.
- Model: A structure used to interpret the operation of a formal system: a structured representation of the real. Models are usually at different scales or levels from the reality they seek to represent.

Further Reading

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LESSONS 22, 23, & 24

THE ROLE OF GEOGRAPHY IN NATIONAL DEVELOPMENT WITH SPECIAL REFERENCE TO KENYA.

Introduction

In the previous lessons, we examined new trends in geographic research and teaching. In particular we looked at the positivist movement and the quantitative revolution, noting how they have influenced the subject matter and methodology of geography. We have also noted the emergence of the welfare approach to geography. This new approach focuses on contemporary problems (hunger, environmental degradation, malnutrition, disease etc) facing society.

In these lectures, we are going to look at the role of Geography in National Development drawing specific examples from Kenya. A fundamental question that we shall try to answer is whether Geography is a discipline or a profession, As we shall see geography can make a contribution to national development in the following areas:

- Education
- Research
- Planning and
- Business.

Objectives

By the end of these lessons, You should be able to:

- a) Mention the major areas in which geography can make a contribution to national development.
- b) Explain the role of geography in Education, research and planning and
- c) Distinguish between geography as a discipline and geography as a profession.

Content.

Question

Is Geography a Discipline or a Profession?

Pursuit of an academic discipline in modern society is part of a career undertaken for financial and other gains. To most of its practitioners the career is a professional one, complete with entry rules and behavioural norms. The academic geographer is distinguished from other geographers (many of whom are also teachers) by their commitment to all three of the basic canons of a University: to propagate, protect, and advance knowledge. It is the advancement of knowledge through the conduct of fundamental research and the publications of its original findings, which identifies an academic discipline. The nature of its teaching follows from the nature of its research.

In the University context a discipline is a branch of learning that involves some method of training/teaching. Several disciplines are taught at the University, (e.g. History, Sociology, Chemistry, etc.). However, each of these disciplines is in one-way or another unique from the other. As indicated earlier Geography is unique from other disciplines taught at the University in the sense that its main focus is on the interaction and

interrelationship of phenomena spatially. The key word here is 'space'. No other discipline stresses on space in its study. Looking at our geography course programs in Geography Department Kenyatta University, one get the impression that many of the course descriptions taught at the department do include the spatial component. To this end one can rightly say that the department is teaching geography as a University discipline just like history, sociology and so on.

A profession, according to the Longman Dictionary of Contemporary English (1984), is a form of employment, especially one that is respected in society as honourable and is possible only for an educated person and after training in some special branch of knowledge. Can geography therefore be a profession?

To be a geographer, one must have an in-built curiosity about the different things that surround us, a love for the differences in the land and man's adjustments to these differences. This is the thread that binds together all geographers who embrace the field for their lifework (Association of American Geographers and the National Council for Geographic Education).

Of course there are several branches of geography. For all of them the same general kinds of interests are needed, but each also requires a somewhat different emphasis in the type of preparation. Some geographers are interested in what makes one place different from another, and why, such must specialize in certain kinds of areal differences. For example they may become interested in how man's natural surroundings differ from place to

place. They know that natural surroundings include such phenomena as the forms and slopes of the land surface, the climate, the water or mineral resources, the plants and animals, and the underlying soils that are made up from a mixture of the inorganic and the organic. It is possible to pursue a career in geography by paying attention to just one of these elements, or by looking at them in combination. In combination they make up the natural surroundings of man, or the human habitat.

Other branches of geography are specifically concerned with man and the different ways in which he carries on his economic, social, and political activities. High density versus scarcity of population, poverty versus wealth of economies, overexploitation versus underexploitation of resources, all these have areal characteristics. Geographers who concern themselves with such comparisons are interested in the spatial element in human behaviour.

No matter what the branch of geography, one can generally identify a geographer by watching what he does. Geographers always want to know where things are located. They look for different kinds of things that are located close together in the same bit of earth surface. For example, they are not only interested in landforms, and soils, farms, and roads considered separately; they are especially interested in the ways such different kinds of things are interconnected. Specific types of farms are found on specific landforms and specific soils. The distance to market or the laws that govern the use of the land affect the location of a farm. Because of the spatial interconnectivity of such elements, a change in any one of them has an effect on all the others. Such a group of

interconnected elements is called a system. Geographers are concerned with the system of interconnected parts and therefore they are spatial system analysts. As spatial system analysts Geographers are professionals in their own right.

According to Darkoh (1992) professional geographers must endeavor to do research and find solutions to issues pertaining to the form of our space-economies, structural dependency, the lack of congruence between the pattern and growth of domestic resource use and the pattern and growth of domestic demand and exploitation of our national territories by governments and businesses of capitalists and socialists nations, drought, desertification, and land degradation, resource use and misuse, pollution, the food crisis, the population crisis, the communication and transportation crisis, the imbalances and disorders of urban and rural life, race-space inequality, poverty and deprivation, social inequality and injustice to name but a few.

The question to ask is – As geography educators are we giving our students the kind of training and professional orientation that includes practical problem-solving situations in order to meet some, if not all these challenges. In other words is our curriculum tailored towards producing students who know Geography as just an academic discipline or as a profession that can provide practical solution to human problems?

Professional geographers

Today, almost every field involving career preparation has its proponents claiming a measure of professionalism. The question of professionalism is one replete with emotion-

laden issues, stemming as often from semantical misinterpretation as from reasoned disagreement. Traditionally a profession is a calling often requiring extensive academic preparation. Today the lists for criteria for a profession are numerous; however the following criteria are common to most listing: an organized body of knowledge, a code of ethics, and a professional culture nurtured by professional associations (Bittel, 1978).

Examining the stated criteria one would agree that geography does not meet these standards, not only to a very small degree.

Having shown that geography is a distinct body of knowledge we shall try to show the involvement of a geographer in dispensing that knowledge in various areas of our national endeavours.

There are many areas where geographers are serving and/or have contributed in Kenya. Such areas include:

- Population movements (e.g. Ominde, 1968)
- Industrilization (e.g. Ogendo, 1972)
- Problems of agriculture an settlement (e.g. Ondingo, 1971)
- Ecological and socio-economic consequences of river damming (e.g. Odingo, 1979).
- Gemorphology work in parts of Rift Valley (e.g. Nyamweru, 1967)
- Regional planning (e.g. Obudho, 1981)
- Soil erosion (e.g.Omwega, 1984, 1989 ; Obando, 1996)
- Transport planning and development (Ogonda,1984; Khayesi,1999)
- Geographic application of remote sensing (Kairu,1989)
- Monitoring of environmental quality (Darkoh,1987, Ogutu1997)

- Energy development (Glowatzki, 2000)
- Urban climatology (Adebayo, 1987, Makokha, 1990)
- Land resource use inventory (Jaetzold, 1982).
- Water policy formulation (Ondieki, 1994).
- Geographical researches touching on the utilization of geothermal energy to produce electricity at Olkaria, Naivasha.
- Agriculture and Food Security (Shisanya, 1996, Wambugu, 2002).

The list is endless. Let us now deal with professional geographers' contribution in Research, Education, Planning, Government agencies and Business.

Professional geographers in Research.

Geographers have been closely attached to development-oriented researches in Kenya in the last four decades (Sindiga, et. al., 1990). Geographers have as individuals (as enumerated above) or with other specialist collaborated in interdisciplinary research into development problems. A geographer functions as “effectively as he can to synthesize the many diverse elements characteristic of geographical projects and form them into an easily understandable whole” (Garofalo and Nicholas, 1984). The geographers effectiveness in such tasks is outlined by Garofalo and Nicholas (AAG, 1984) as a result of:

- (i) Broad academic experience in the physical, biological and social sciences.
- (ii) Knowledge of various tools and methods used for data collection and analysis, map preparation and reproduction.
- (iii) Interest in the interrelationships among sciences and;

- (iv) Interest in coordinating the effect of professionals from various disciplines to achieve project goals.

Although geography is a prominent academic discipline at the University, what Anderson a geographer of AAG, 1979 wrote is certainly applicable in Kenya today:

“...Geography has gained in academic stature at the University (but) it is unfortunate to note that many pressing problems facing our nation and needing the rigorous application of geographic knowledge are not being directed to professional geographers in an interdisciplinary project for them to analyze and arrive at satisfactory solutions (Anderson, 1979).

Furthermore, scholarly ranks are too small to cope with enormous research agenda. Research opportunities are limited, partly by inadequate funding (in addition to increasing teaching workloads). Also intellectual and professional nourishment is limited by lack of outlets of research publication in the Eastern and Southern Africa region (Sindiga, et. al., 1990).

Activity

Citing specific examples outline the role of geographers in research.

Professional geographer in Education

Let's briefly examine the professional geographers' contribution in the following education areas: University teaching, Administration, examinations, writing of textbooks and also non-formal education.

(i) University teaching.

Aided by sophisticated analytical tools of contemporary social science, along with other traditional geographic tools, geography is increasingly covering a diversified field of

learning and teaching. Modern geographers are preying and gaining insight into the problems the contemporary world – problems of urbanization, population growth and distribution, famine and food supply, transportation, housing, resource use and misuse, quality of man’s physical and social environment etc, and as such it is increasingly becoming difficult to draw a boundary line between geography and related disciplines such as economics, political science, ecology and sociology (Darkoh, 1977).

The disheartening thing is (and unfortunately true) “... geography students will graduate without acquiring a sense of professional self awareness and self identity. This gloomy prospect is as a consequence of lack of attention that we educators give to satisfying our student’s need for such a perspective” (Meir, 1982). To establish a professional self-identity within the students the educators must view it as being similar to the human personality. This means that undergraduates should be taught history and philosophy of geography as a compulsory rather than an elective course, so as to enable them to relate as members of the geographic community and to geography as part of the larger scientific community. Of course this is a difficult task taking into consideration the heavy teaching loads of the staff, and the length of time during which they are exposed to geography education (in comparison to the time available for graduate students). All the same the picture is not as grim as one would tend to quickly conclude; since the University is providing training to graduates in many areas that are needed in resource surveys, thus transferring not only knowledge but also technology from the “towers” to the society.

(ii) Geographers in Administration and in Examinations

Many geographers in addition to classroom teaching have been called to play administrative roles at different levels in our education system. For example as heads of institutions, members of Kenya Institute of education subject panels producing course syllabi (Curriculum developers), examiners and administrators in national examinations, etc.

(iii) Geographers and Writing of textbooks

Many professional geographers as teachers in primary schools, secondary schools, University and in specialized educational professional areas came together in 1984 after the implementation of the 8.4.4. Education system to rewrite a relevant syllabus and textbooks for all classes in secondary and primary schools. Notable among those who answered to the call of relevant textbooks are:

- Nyamweru – a series of books for primary school geography
- Wegulo & Macmaster – secondary school map work
- Aluko Orodho – secondary school series of fundamental geography
- Watson, J. – physical geography for secondary schools.
- K.I.E. Geography – geography text series for secondary schools (authored by a team of teachers).

Probably the most famous geography team contribution was in the preparation of the Kenya National Atlas (1970). Leading personalities with geographical training from the Survey of Kenya; University College, Nairobi; E.A. Meteorological department; Ministry of Agriculture; Lands department, Forestry; Department of mines and geology among

others, contributed to this useful information volume with a world wide demand from Libraries, Universities and centres of information and research.

(iv) Geographers and Non-formal education

Geographers can also contribute to the welfare of the society by transmitting their knowledge outside the formal school system. This is when they join wananchi in such areas as soil erosion campaigns, National tree planting days, Public Barazas to discuss important issues such as energy crisis, new technologies in agriculture, family planning, etc. Also during Agricultural trade fairs geographers play an important part in interpreting the themes for various stands, in addition to educating the masses through such involvement.

Professional geographers in planning and government agencies

Salter (1983) in an article “what can I do with geography?” Noted that for the individual, whose professional life is devoted to teaching and research, this question embraces the understanding of people, systems, and places of the earth. This question thus functions as a catalyst for much personal and professional growth. Many geographers are engaged in research within government agencies and other bodies. These geographers serve with other position titles other than that of geographers, although the work they are doing is quite geographical. It would be helpful in achieving more widespread acceptance of geography as a discipline if geographers were clearly identified as geographers in government services. As this is most unlikely to happen then, professional geographers should maintain their identity with the discipline through active participation of geographic seminars, conferences, journal contributions in addition to speaking out in a

geographical perspective in the context of their every day work activities (Anderson, AAG, 1979).

It is probably interesting to note that a prominent geographer was appointed to chair the first commission to deliberate on the goals of education in an independent Kenya. This was Professor S.H. Ominde in 1964.

Professional geographers in Business

Geographers have advanced to management positions within companies and have proven to be especially effective in the coordination of projects requiring the interaction of scientists and technicians with different background and skills. AAG (1984) noted that over the past decade the hiring of geographers in the middle and management positions had risen from 20% to 40% in the U.S. and this is a indication of he fact that geographers are increasingly being relied upon to provide direction as well as technical expertise. Geographers as manages recognize the need for specialized knowledge in other fields and have been effective at directing this specialized knowledge appropriately to successful project completion (Garofalo and Nicholas, 1984).

Conclusion

Geographers are very well placed to diagnose and prescribe antidotes for most of our contemporary problems. The breath of their training and their ability to handle and synthesize materials from a range of sources; their acceptance of the complexity of causation; the range of information which they are trained to tap, their interest in

distributions, and their long tradition of study in this area all point to the propensity of potential the geographer is equipped with to handle contemporary problems.

In conclusion it is our considered opinion that time is ripe when geography curriculum should balance between training geographers as academicians (i.e. geography teachers) and as professionals, such as cartographers, Businessmen, surveyers, environmentalists, climatologists, geomorphologists, agricultural geographers, urban planners, regional planners, resource evaluators, industrial location geographers, marketing and transport geographers, G.I.S. and Remote sensing experts, etc.).

Are we as academic geographers preparing for posterity a foundation of geography that will ensure man understands, manages and organizes space in an optimal manner through time? Shall our students make this a better world?

Summary

In these lessons we have looked at the role of geography in national development. A fundamental question we have tried to address is whether geography is an academic discipline or a profession. We have noted that geographers can make a contribution to national development in a number of areas namely education, research, Planning and in business.

ACTIVITY

Describe the role of geography in national development
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Definition of Key words

- Discipline: A branch of learning that encompasses some unique method of training/teaching.
- Profession: A form of employment, especially one that is respected in society as honourable and is possible only for an educated person and after training in some special branch of knowledge.
- Science: In general usage the term refers to knowledge and particularly to systematized knowledge. The scientific method is widely seen as involving the construction of theories, which seek to explain, observed events by reference to unobserved forces.

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